ORIUS Professor Appendix A: Formal and Algorithmic Surface

Author details omitted for professor review draft

I. Appendix Overview

This appendix collects the formal and algorithmic material that sits behind the professor-facing ORIUS IEEE draft: notation, assumptions, theorem statements, proof maps, adapter semantics, and the strongest theorem-side surfaces reused from the canonical dissertation. The appendix is intentionally heavier than a venue supplement. Its role is to make every strong formal claim in the main paper traceable to a local runtime object, an explicit assumption surface, or a proof-facing artifact.

II. Appendix A: Formal Notation, Assumptions, Theorem Register, and Proof Map

This appendix is the compact formal reference for the professor-facing IEEE draft. It collects the notation, the standing assumptions, the theorem register, and the proof-map framing used in the main theory bridge. The goal is to keep the body readable while making every strong claim traceable to a formal object or a local source file.

A. Notation

B. Standing assumptions

The formal objects used in the body are grounded in the assumption register from the local thesis sources. The most important assumptions are:

- **A1: bounded model error.** The nominal dynamics error is bounded.
- **A2: bounded telemetry error.** Observation error is controlled by a monotone function of reliability.
- **A3: feasible safe repair.** If a safe state exists, the tightened action set is non-empty.
- **A4: known dynamics.** The local transition law is available to the controller or safety layer.
- **A5: monotone bounded inflation.** Lower reliability never shrinks the uncertainty surface.
- **A6: bounded detector lag.** Degradation is detected within a finite lag.
- **A7: causal certificate rule.** Certificates depend only on current and past information.
- **A8: admissible fallback.** A safe fallback policy exists under the admissible fault model.

These assumptions are not decorative. They are the minimum conditions under which the theory bridge can claim that the universal runtime contract is not just readable, but sound.

C. Theorem register

D. Proof-map framing

The proof map used in the main text has three levels.

1) **Reference-domain proofs.** The battery ladder and the supporting theorem surface establish the reference proof objects. These are the source of the concrete runtime semantics and the strongest empirical closure.
2) **Transfer proofs.** The universality chapter isolates the objects that a new domain must supply to inherit the same one-step safety statement.
3) **Promotion proofs.** The cross-domain synthesis and deployment sections use the universality gate to decide whether a domain may be treated as defended, bounded, gated, or experimental in the article body.

In practice, the proof map is read as a dependency chain:

$$\text{A1–A8} \Longrightarrow \text{battery reference ladder} \Longrightarrow \text{T9/T10/T11} \Longrightarrow \text{universality gate} \Longrightarrow \text{domain promotion status.}$$

This is the simplest way to keep the professor-facing draft universal-first without collapsing evidence tiers. The theorem register tells the reader what is proved, and the proof map tells the reader what has to be true before those proofs can be reused.

E. How to use this appendix

When a claim in the main IEEE text depends on a formal step, cite the theorem tag from Table II and the assumption ID from the register above. When a claim depends on a battery-specific derivation, use the proof-map framing to make clear whether the claim is a reference-domain result, a transfer theorem, or a promotion-level statement. This avoids the common failure mode in stitched monographs: letting broad architectural language outrun the actual proof obligations.

TABLE I: Canonical notation for the ORIUS professor draft.

Symbol / Term	Meaning
x_t	True plant state.
y_t	Telemetry or observation at time t.
$\mathcal{H}_t$	Observation history generated by $y_{0:t}$.
$w_t \in [0,1]$	Reliability score extracted from telemetry history.
U_t	Observation-consistent state set.
$\mathcal{S}$	Safe state set defined by the domain safety predicate.
$\mathcal{A}_t^{\text{safe}}(U_t)$	Tightened safe-action set under uncertainty set U_t.
a_t^{cand}	Nominal candidate action produced by the upstream controller.
a_t^{safe}	Repaired or fallback action emitted by ORIUS.
V_T	Episode-level true-state violation count.
α	Miscoverage or per-step risk budget.
$\bar{w}$	Mean reliability across the horizon.
$\phi(x)$	Safety predicate defining the domain safe set.

TABLE II: Professor-facing theorem register for the ORIUS draft.

Tag	Claim	Role in the paper
T9	Reliability-aware repair is structurally necessary.	Establishes that ignoring w_t yields linear violation scaling under persistent degradation.
T10	Degraded observation induces a reliability-risk frontier.	Lower bound showing that the risk envelope cannot be removed by controller choice alone.
T11	Structural transfer through the universal adapter contract.	Shows that the battery proof pattern transfers when a domain supplies the required runtime objects.

III. Assumptions, Notation, and Proof Discipline

This chapter establishes the formal scaffolding for Part III. It fixes notation that is stable across all theorems, catalogues the eight standing assumptions, defines the proof dependency matrix, and sets the disciplinary standards that every subsequent proof must satisfy. The point of the chapter is not to enlarge the formal inventory for its own sake; it is to make the later results readable as one coherent theorem program. The flagship DC3S ladder in Chapters IV–VIII contains the battery reference theorems (T1–T8), but the full dissertation surface is broader: Part I contributes the precursor results, the later chapters add lemmas, propositions, and corollaries, and the monograph extension closes with T9–T11. Counting the active master-manuscript include tree yields **83 formal items** (26 theorems, 5 lemmas, 9 propositions, 12 corollaries, 15 definitions, 16 assumptions), all audited in Appendix X and summarised in Table VIII at the end of this chapter.

A. Stable Notation

Throughout Part III, the following symbols have fixed meaning:

TABLE III: DC3S Theorem Dependency and Proof Completeness. Each formal result is implemented in code and tested by at least one automated test.

Result	Name	Depends On	Proof Method	Code Reference
Defn. 1	OQE reliability weight w_t	—	Definition	`quality.py:compute_reliability()`
Prop. 1	Inflation monotonicity	Defn. 1	Analytic	`coverage_theorem.py:verify_inflation_geq_one()`
Thm. 1	Marginal coverage guarantee	Prop. 1, Defn. 1	Conformal CP theory	`coverage_theorem.py:assert_coverage_guarantee()`
Thm. 3	Expected violation bound	Thm. 1	Expectation calculus	`coverage_theorem.py:compute_expected_violation_bou`
Thm. 9	Group-conditional coverage	Thm. 1	Mondrian CP	`coverage_theorem.py:mondrian_group_coverage()`
Thm. 10	Hoeffding high-prob. bound	Thm. 3	Hoeffding inequality	`coverage_theorem.py:hoeffding_violation_bound()`
Thm. 11	Byzantine OQE reliability bound	Defn. 1	Trimmed-mean theory	`quality.py:compute_reliability_robust()`

All results are unit-tested in `tests/test_conditional_coverage.py` and `tests/test_adversarial_faults.py`.

Symbol	Meaning
x_t	True (plant-internal) state at step t
$\hat{x}_t$	Observed (telemetry-delivered) state at step t
a_t	Action (dispatch decision) at step t
a_t^{cand}	Candidate action proposed from the observation history
a_t^{safe}	Repaired action returned by the ORIUS kernel
U_t	Observation-consistent state set built from degraded telemetry
$\mathcal{S}$	Safe state set: $\mathcal{S} = \{x : \phi(x) \geq 0\}$
$\phi(x_t)$	Safety predicate: $\phi(x_t) = 1$ iff x_t satisfies all constraints
w_t	Reliability score, $w_t \in [0, 1]$
α	Conformal miscoverage level (nominal $\alpha = 0.10$)
q_t	Conformal quantile (half-width of the prediction interval)
$\mathcal{A}_t$	Tightened safe-action set at step t
$\mathcal{H}_t$	Causal observation history available at time t
T	Episode horizon (number of steps)
V	Total violation count: $V = \sum_{t=1}^{T} \mathbf{1}[\neg\, \phi(x_t)]$
$\bar{w}$	Episode-average reliability: $\bar{w} = T^{-1} \sum_t w_t$

a) Semantics of the reliability score.: The symbol w_t is treated throughout this dissertation as a *runtime reliability score*, not as a probability by definition. It records how trustworthy the telemetry channel appears at step t according to the OQE pipeline and adjacent runtime checks. A theorem may convert w_t into a probabilistic statement only when that conversion is spelled out explicitly as an extra theorem-local hypothesis. In particular, later chapter statements do not identify w_t with the probability of clean telemetry unless the statement itself introduces a calibration or indistinguishability contract that justifies that move.

b) Battery instantiation.: In the battery domain, $x_t = \text{SOC}_t$ (MWh), $a_t = (P_t^{\text{chg}}, P_t^{\text{dis}})$ (MW), and $\phi(x_t) = \mathbf{1}[\text{SOC}_{\min} \leq x_t \leq \text{SOC}_{\max}]$.

B. Assumption IDs (A1–A8)

Each assumption is assigned a permanent identifier and stated in both domain-general and battery-specific form.

Assumption 1 (A1: Bounded model error)**.** The one-step prediction error $\|x_{t+1} - f(x_t, a_t)\|$ is bounded by a known constant ϵ_{model}. *Battery:* $|x_{t+1} - (x_t + \eta\, P_t\, \Delta t)| \leq \epsilon_{\text{model}}$.

Assumption 2 (A2: Bounded telemetry error)**.** The observation error $\|\hat{x}_t - x_t\|$ is bounded by a reliability-dependent function: $\|\hat{x}_t - x_t\| \leq g(w_t)$, where g is monotone decreasing.

Assumption 3 (A3: Feasible safe repair)**.** For every state x_t satisfying $\phi(x_t) = 1$, there exists at least one action $a_t \in \mathcal{A}_t$ such that $\phi(x_{t+1}) = 1$.

Assumption 4 (A4: Known dynamics)**.** The transition function $f(x_t, a_t)$ is known to the controller up to the model error bounded in A1. *Battery:* the SOC transition $x_{t+1} = x_t + \eta\, P_t\, \Delta t$ is known with efficiency η identified offline.

Assumption 5 (A5: Monotone bounded uncertainty inflation)**.** The inflation function $\iota(w_t) = q_t/(w_t + \epsilon)$ is monotone decreasing in w_t and bounded: $\iota(w_t) \leq \bar{\iota}$ for all $w_t \in [0, 1]$.

Assumption 6 (A5′: Calibration regularity)**.** The telemetry error bound $g(w_t)$ from Assumption A2 satisfies $g(1) = 0$ and is Lipschitz near $w_t = 1$: there exist constants $c_g > 0$ and $\delta_w \in (0, 1)$ such that $g(w_t) \leq c_g\,(1 - w_t)$ for all $w_t \in [1 - \delta_w, 1]$. This calibration regularity aligns the error bound with the reliability score and is used in the information-theoretic lower bound (T10).

Assumption 7 (A6: Bounded detector lag)**.** The OQE and adjacent runtime detectors recognize supported telemetry faults within channel-specific finite lags. The active detector-lag contract is

$$\tau_{\max}(\text{dropout}) = 1,\ \tau_{\max}(\text{stale}) = 3,\ \tau_{\max}(\text{delay}) = 2,\ \tau_{\max}(\text{OOO}) = 10,\ \tau_{\max}(\text{spike}) = 1,\ \tau_{\max}(\text{drift}) = 48.$$

If a fault begins at step t_f, then the corresponding reliability penalty or dedicated detector flag must fire by step $t_f + \tau_{\max}$.

Assumption 8 (A6′: φ-mixing fault process)**.** The fault injection process $\{F_t\}$ is φ-mixing with geometric decay: there exist constants $C_\phi > 0$ and $\rho \in (0, 1)$ such that the φ-mixing coefficient satisfies $\varphi(k) \leq C_\phi\, \rho^k$ for all $k \geq 0$. A buffer of $\tau_{\text{mix}} = \lceil \log(1/\varepsilon)/|\log \rho| \rceil$ steps makes any two windows ε-approximately independent in total variation. Assumption A6′

is used in the universal impossibility proof (T9) and is satisfied by any i.i.d. or geometrically ergodic Markov fault model.

Assumption 9 (A7: Causal certificate update)**.** The certificate $\mathcal{C}_t$ depends only on information available at or before step t: $\mathcal{C}_t = h(\hat{x}_{1:t}, a_{1:t-1})$.

Assumption 10 (A8: Admissible fallback policy)**.** There exists a fallback policy π_{fb} that, starting from any safe state, maintains $\phi(x_t) = 1$ for all future steps under worst-case bounded disturbance. *Battery:* the zero-dispatch policy ($P_t = 0$) is always admissible because SOC is unchanged (up to model error).

a) Executable universal kernel.: These assumptions are not merely declarative. Their runtime realization is concentrated in the typed ORIUS kernel, whose implementation surface records observation packets, reliability assessments, observation-consistent state sets, safe-action sets, repair decisions, and safety certificates. The battery-domain proofs remain the reference instantiation, but the reusable universal claim now attaches to that typed kernel rather than to an informal adapter narrative.

TABLE IV: Assumption-to-code traceability for the universal degraded-observation kernel.

ID	Assumption	Canonical code surface	Runtime status
A1	Bounded model error	Domain instantiation and battery reference helpers in `src/orius/universal_theory/battery_instantiation.py`	Declared by model/artifact surface
A2	Bounded telemetry error	`compute_oqe()`, `build_reliability_assessment()`, observation-consistent state construction	Checked through reliability and state envelopes
A3	Feasible safe repair	`tighten_action_set()`, `repair_action()`, `build_repair_decision()`	Checked per step
A4	Known dynamics	`DomainInstantiation` contract and domain-specific transition models	Declared by domain instantiation
A5	Monotone bounded inflation	`verify_inflation_geq_one()` and clipped inflation in `build_observation_consistent_state_set()`	Checked per step
A6	Bounded detector lag	OQE/drift artifacts and locked replay audits	Artifact-validated; not fully checked at runtime
A7	Causal certificate update	`execute_universal_step()` and `SafetyCertificate.from_payload()`	Checked structurally
A8	Admissible fallback policy	`SafetySpec.fallback_action`, battery zero-dispatch fallback, fallback branch in `build_repair_decision()`	Checked when fallback exists

C. Proof Dependency Matrix

Table V shows which assumptions each theorem requires.

TABLE V: Assumption and local-contract dependencies of the full T1–T11 theorem program.

Theorem	A1	A2	A3	A4	A5	A5′	A6	A6′	A7	A8	Extra local contract or witness
T1 OASG Existence	✓	✓		✓							Boundary-reachability witness near the constraint surface
T2 Safety Preservation	✓	✓	✓	✓	✓				✓		Inflated-set containment plus repair membership in the tightened safe set
T3 Core Bound	✓	✓		✓	✓		✓		✓		Predictable risk-envelope budget $r_t \leq \alpha(1 - w_t)$
T4 No Free Safety		✓									Fixed-margin, quality-ignorant controller class and admissible degraded-observation witness
T5 Certificate Horizon	✓	✓		✓	✓		✓		✓		Forward-tube containment construction
T6 Certificate Expiration Bound				✓	✓		✓		✓		Drift-scale lower-bound argument for certificate expiration
T7 Feasible Fallback Existence	✓		✓	✓						✓	Interior-state fallback construction
T8 Graceful Degradation Dominance	✓	✓	✓	✓	✓					✓	Shared-fault comparison coupling between graceful and uncontrolled release
T9 Universal Impossibility				✓				✓			Per-window degraded-observation witness under persistent degradation
T10 Boundary-Indistinguishability Lower Bound	✓	✓				✓					Boundary prior and observation-law indistinguishability hypothesis
T11 Typed Structural Transfer											Coverage, soundness, repair-membership, and fallback-admissibility obligations

Reading note. The first ten columns record standing assumptions. The final column records theorem-local contracts or structural witnesses that are not promoted to global assumptions because they are only needed for the specific theorem in that row.

1) Extended Monograph Theorem Program: The flagship dependency matrix above covers the battery proof ladder. The monograph-level extension adds T9–T11, each of which answers a distinct control-theory question and introduces an explicit new obligation.

TABLE VI: Extended theorem program beyond the battery ladder.

Theorem	Question answered	New assumptions	Role
T9	Is reliability-aware repair structurally necessary under persistent degradation?	A6′ + boundary reachability witness	Necessity theorem for degraded-observation control
T10	When does degraded observation force a nontrivial reliability-risk frontier?	Explicit boundary prior and observation indistinguishability assumptions	Stylized lower-bound frontier theorem
T11	Which structural obligations let the battery safety argument transfer to a new domain?	Explicit soundness, repair-membership, and fallback obligations	One-step structural transfer theorem

2) Gap-Closure Map: The theorem program is easier to audit if each thesis-level gap claim is tied to the theorem that closes it and to the residual boundary that still remains. Table VII makes that dependency explicit.

TABLE VII: From gap statement to theorem closure and current claim boundary.

Gap statement	Closing result	Formal object introduced	Current status
Observed-state safety can disagree with true-state safety.	S1, T1, T2	Observation-consistent state set and repaired safe action	Closed on the battery reference surface.
Ignoring reliability may still look operationally attractive.	T4, T9	Necessity of reliability-aware repair under persistent degradation	Closed as a structural impossibility result; battery remains the witness surface.
The $\alpha(1-\bar{w})T$ envelope may be only a loose upper bound.	T3, T10	Reliability-risk frontier under explicit boundary-indistinguishability assumptions	Closed conservatively on a stylized frontier model; full sharpness remains battery-anchored.
Cross-domain portability could be mistaken for equal proof depth.	T11 + tier tables	Explicit one-step transfer obligations and tiered discharge rules	Closed at the contract level; empirical discharge remains tiered by domain.
The AV residual could be misread as a contradiction of the theory.	Universality-completeness chapter (constraint-class mismatch analysis)	Constraint-class mismatch diagnosis	Open promotion gap, but not a contradiction of T1–T11.
Navigation and aerospace rows could be over-read as proof peers.	Claim-boundary discipline in Chapter I	Evidence-tier interpretation	Intentionally outside the proof-validating surface.

D. Marginal Versus Conditional Coverage

A critical distinction throughout Part III:

Marginal coverage. $\Pr[\, x_t \in \hat{C}_t \,] \geq 1 - \alpha$ averaged over the randomness of calibration and test points. This is what CQR provides.

Conditional coverage. $\Pr[\, x_t \in \hat{C}_t \mid \hat{x}_t \,] \geq 1 - \alpha$ for every observed value. Conditional coverage is strictly stronger and is *not* guaranteed by CQR without additional distributional assumptions.

The ORIUS theorems are stated under marginal coverage unless a stronger conditioning statement is written explicitly into the theorem assumptions. The battery reference ladder uses monotone inflation to preserve the base coverage surface. The universal transfer theorem (T11) is kept at the one-step level for exactly this reason: it does not silently upgrade marginal coverage into a stronger episode-level conditioning claim.

E. Theorem-Local Calibration Contracts

Two theorem-local contracts are used repeatedly in the tightened theory surface:

1) **Risk-envelope contract.** A chapter may introduce a predictable sequence r_t satisfying $\mathbb{P}[x_{t+1} \notin \mathcal{S} \mid \mathcal{H}_t] \leq r_t$. This is the correct bridge from one-step conditional reasoning to an episode-level expectation bound. The battery-style envelope $r_t \leq \alpha(1 - w_t)$ is one such contract, but it is not claimed to follow from vanilla conformal exchangeability alone.
2) **Boundary-indistinguishability contract.** A chapter may model degraded observation through pairs of latent hypotheses whose observation laws remain close in total variation or another explicit channel metric. This is the extra ingredient used in the stylized lower-bound theorem (T10); it is not inherited automatically from the definition of w_t.

These contracts are local on purpose. They keep the theorem program honest: coverage language stays marginal unless a stronger conditioning statement is written into the theorem, and reliability language stays operational unless a calibration or indistinguishability hypothesis is stated alongside it.

F. One-Step Versus Multi-Step Claims

The theorems in Chapters IV–VI operate on a *per-step* basis: each step's safety is analysed conditionally on the current state. The core bound (Chapter VI) aggregates per-step risks into an episode-level expectation via linearity of expectation.

Multi-step claims (certificate horizon, graceful degradation, and any episode-level transfer bound) require additional assumptions about temporal dependence or explicit per-step risk budgets; these are stated explicitly in Chapter VIII and in the aggregation corollaries that follow the relevant one-step theorems.

G. The Theorem Ladder

The theorems form a logical ladder:

1) **OASG Existence** (Chapter IV): The problem is real—observed safety does not imply true safety.
2) **Safety Preservation** (Chapter V): The shield solves the problem—true safety is restored.
3) **Core Bound** (Chapter VI): The solution has a quantifiable cost—expected violations are bounded.
4) **No Free Safety** (Chapter VII): The cost is unavoidable—ignoring observation quality breaks safety.
5) **Temporal Extensions** (Chapter VIII): The guarantee extends across time—certificates expire gracefully.

Each rung depends on the ones below it, and no rung can be removed without breaking the chain.

H. Theorem-Writing Discipline

Every theorem in this dissertation follows a fixed template:

1) **Statement**: formal claim with quantifiers.
2) **Assumptions used**: list of A-IDs.
3) **Proof**: complete derivation.
4) **Artifact link**: pointer to the empirical evidence that instantiates the theorem in the battery domain.
5) **Scope disclaimer**: "What this theorem does not prove."

I. Common Theorem Failure Modes

To inoculate against reviewer objections, we catalogue failure modes that a careless proof might exhibit:

1) *Vacuous truth.* If the assumption set is never satisfied, the theorem is true but useless. We verify non-vacuousness by exhibiting the battery instantiation.
2) *Circular dependence.* Theorem X assumes the conclusion of Theorem Y, and vice versa. The dependency matrix (Table V) prevents this.
3) *Hidden conditional.* A marginal claim is stated as if it were conditional. We use the marginal/conditional distinction explicitly.
4) *Infinite-sample limit.* A result that holds only asymptotically is presented as finite-sample. All ORIUS bounds are stated for finite T.

J. Integrated Theorem Surface Summary

Table VIII enumerates the full formal surface of the dissertation. That count now includes the foundational precursor results (S1–S2), the battery proof ladder (T1–T8), and the monograph extension theorems (T9–T11). The integrated theorem gate still enforces the same minimum standard for every item: source file exists, code anchor verified, artifact locked when applicable, and appendix mapping confirmed.

TABLE VIII: Integrated theorem-like surface counts generated from the active thesis include tree.

Environment	**Count**
Theorem	26
Lemma	5
Proposition	9
Corollary	12
Definition	15
Assumption	16
Total	**83**

K. Chapter Summary

This chapter provides the stable notational, assumptive, and methodological foundation for every proof in Part III. The eight assumptions (A1–A8) are individually justified in both domain-general and battery-specific terms, the proof dependency matrix prevents circular reasoning, and the theorem-writing discipline ensures that every claim is bounded by explicit scope disclaimers. The integrated theorem surface of 83 formal items—spanning 26 theorems, 5 lemmas, 9 propositions, 12 corollaries, 15 definitions, and 16 assumptions—is the formal depth this dissertation contributes to the CPS safety literature.

IV. Theorem 1: OASG Existence

This chapter proves the *Observed-safe / Actually-unsafe State Gap* (OASG) existence theorem in the battery domain. The theorem establishes that telemetry degradation creates episodes where an action is safe according to the observed state but unsafe according to the true state—the foundational phenomenon that motivates the entire ORIUS safety layer.

A. Domain-Agnostic Statement

We first state the OASG existence result for an arbitrary cyber-physical system; the battery domain provides the concrete instantiation that follows.

Definition IV.1 (Abstract CPS)**.** A cyber-physical system is a tuple $(\mathcal{X}, \mathcal{U}, f, \mathcal{S}, \hat{x})$ where $\mathcal{X} \subseteq \mathbb{R}^n$ is the state space, $\mathcal{U} \subseteq \mathbb{R}^m$ is the action space, $f : \mathcal{X} \times \mathcal{U} \to \mathcal{X}$ is the (possibly stochastic) dynamics, $\mathcal{S} \subseteq \mathcal{X}$ is the safe set with non-empty boundary $\partial\mathcal{S} \neq \emptyset$, and $\hat{x}_t = x_t + \xi_t$ is the observed state under telemetry error ξ_t with $\|\xi_t\| \leq g(w_t)$ (Assumption A2).

Theorem IV.2 (OASG Existence — General CPS)**.** *Let $(\mathcal{X}, \mathcal{U}, f, \mathcal{S}, \hat{x})$ be a CPS satisfying Assumptions A1 (bounded dynamics error), A2 (bounded telemetry error with $g(w_t) > 0$ under degraded quality), and A4 (known dynamics). Let $\pi : \mathcal{X} \to \mathcal{U}$ be any quality-ignorant controller, i.e. one whose action depends only on $\hat{x}_t$ and not on w_t.*

If the controller drives the state within distance $\delta > 0$ of $\partial\mathcal{S}$ with positive frequency, and the telemetry error satisfies $\Pr[\|\xi_t\| > \delta] > 0$, then there exists a time t^* such that

$$\phi_{obs}(\pi(\hat{x}_{t^*})) = 1 \quad \textit{and} \quad \phi_{true}(\pi(\hat{x}_{t^*})) = 0,$$

where ϕ_{obs} and ϕ_{true} are the observed-safe and truly-safe predicates respectively.

The proof constructs the witness t^* by intersecting the boundary-proximity set (from optimal control) with the observation-error tail (from degraded telemetry). We now instantiate this in the battery domain.

B. Battery-Domain Instantiation

Consider a battery energy storage system with:

- True state $x_t = \mathrm{SOC}_t \in [0, E_{\max}]$ (MWh).
- Observed state $\hat{x}_t = x_t + \xi_t$, where ξ_t is the telemetry error induced by packet dropout, delay, drift, or stale sensors.
- Action $a_t = (P_t^{\mathrm{chg}}, P_t^{\mathrm{dis}}) \in \mathbb{R}_+^2$ subject to power and ramp constraints.
- Dynamics: $x_{t+1} = x_t + (\eta_c\, P_t^{\mathrm{chg}} - \eta_d^{-1}\, P_t^{\mathrm{dis}})\, \Delta t + \epsilon_t$, where $|\epsilon_t| \leq \epsilon_{\mathrm{model}}$ (Assumption A1).

C. Observed and True Safety Predicates

Definition IV.3 (Observed safety)**.** An action a_t is *observed-safe* if the predicted next state under the observed SOC satisfies the constraints:

$$\phi_{\mathrm{obs}}(a_t) \;=\; \mathbf{1}\big[\,\mathrm{SOC}_{\min} \;\leq\; \hat{x}_t + \Delta(a_t) \;\leq\; \mathrm{SOC}_{\max}\,\big],$$

where $\Delta(a_t) = (\eta_c P_t^{\mathrm{chg}} - \eta_d^{-1} P_t^{\mathrm{dis}})\, \Delta t$.

Definition IV.4 (True safety)**.** An action a_t is *truly safe* if the actual next state satisfies the constraints:

$$\phi_{\text{true}}(a_t) \;=\; \mathbf{1}\big[\,\text{SOC}_{\min} \;\le\; x_t + \Delta(a_t) + \epsilon_t \;\le\; \text{SOC}_{\max}\,\big].$$

The OASG phenomenon occurs when $\phi_{\text{obs}}(a_t) = 1$ but $\phi_{\text{true}}(a_t) = 0$.

D. Assumptions Used

The OASG existence theorem requires:

- **A1** (bounded model error): ensures the dynamics are predictable up to ϵ_{model}.
- **A2** (bounded telemetry error): ensures the observation gap $|\xi_t| \le g(w_t)$ is finite but non-zero under degraded telemetry.
- **A4** (known dynamics): ensures the controller can compute $\Delta(a_t)$ correctly.

E. Auxiliary Lemmas

Lemma IV.5 (Observation gap under dropout)**.** *Under packet dropout with rate $d > 0$, the observation error satisfies*

$$\mathbb{E}\big[|\xi_t|\big] \;\ge\; d \cdot \sigma_{drift} \cdot \sqrt{\Delta t},$$

where σ_{drift} is the standard deviation of the one-step state drift.

Proof. When a packet is dropped (probability d), the observer uses the last received state, which drifts by at least $\sigma_{\text{drift}}\sqrt{\Delta t}$ in expectation under the sub-Gaussian drift model. The result follows by the law of total expectation. □

Lemma IV.6 (Boundary proximity under arbitrage)**.** *An optimal arbitrage controller drives SOC within $\delta_{bnd} = P_{\max}\,\Delta t$ of at least one constraint bound in at least a fraction $f_{bnd} > 0$ of time steps.*

Proof. Optimal arbitrage charges to near-full when prices are low and discharges to near-empty when prices are high. The fraction f_{bnd} depends on the price distribution and is strictly positive whenever price variance is non-zero. □

F. Main Theorem: OASG Existence

Theorem IV.7 (OASG Existence)**.** *Under Assumptions A1, A2, and A4, for any dropout rate $d > 0$ and any optimal arbitrage controller, there exists a time step t^* such that*

$$\phi_{obs}(a_{t^*}) = 1 \quad \textit{and} \quad \phi_{true}(a_{t^*}) = 0.$$

That is, the controller selects an action that appears safe in the observed state but violates a constraint in the true state.

Proof. By Lemma IV.6, there exists a set $\mathcal{T}_{\text{bnd}} \subset \{1, \ldots, T\}$ with $|\mathcal{T}_{\text{bnd}}| \ge f_{\text{bnd}}\,T$ such that for $t \in \mathcal{T}_{\text{bnd}}$, the true SOC is within δ_{bnd} of a constraint bound.

By Lemma IV.5, for dropout rate $d > 0$ the observation error has $\mathbb{E}[|\xi_t|] > 0$. In particular, with positive probability, $|\xi_t| > \delta_{\text{bnd}}$ for some $t \in \mathcal{T}_{\text{bnd}}$.

At such a step t^*, the observed state $\hat{x}_{t^*} = x_{t^*} + \xi_{t^*}$ is displaced from the true state by more than the distance to the constraint boundary. The controller, which sees only $\hat{x}_{t^*}$, selects an action that is feasible under the observed state but pushes the true state across the bound:

$$x_{t^*} + \Delta(a_{t^*}) < \text{SOC}_{\min} \quad \text{while} \quad \hat{x}_{t^*} + \Delta(a_{t^*}) \ge \text{SOC}_{\min}.$$

Hence $\phi_{\text{obs}}(a_{t^*}) = 1$ and $\phi_{\text{true}}(a_{t^*}) = 0$. □

G. Interpretive Table

Table IX maps the theorem's formal elements to battery observables.

TABLE IX: Mapping OASG Existence theorem to battery domain.

Formal element	**Battery meaning**
x_t	True SOC (MWh)
$\hat{x}_t$	BMS-reported SOC after telemetry degradation
ξ_t	Telemetry error from dropout / drift / delay
$\phi_{\text{obs}} = 1, \phi_{\text{true}} = 0$	Controller thinks SOC is in bounds; it is not
δ_{bnd}	Distance from SOC to nearest rated limit

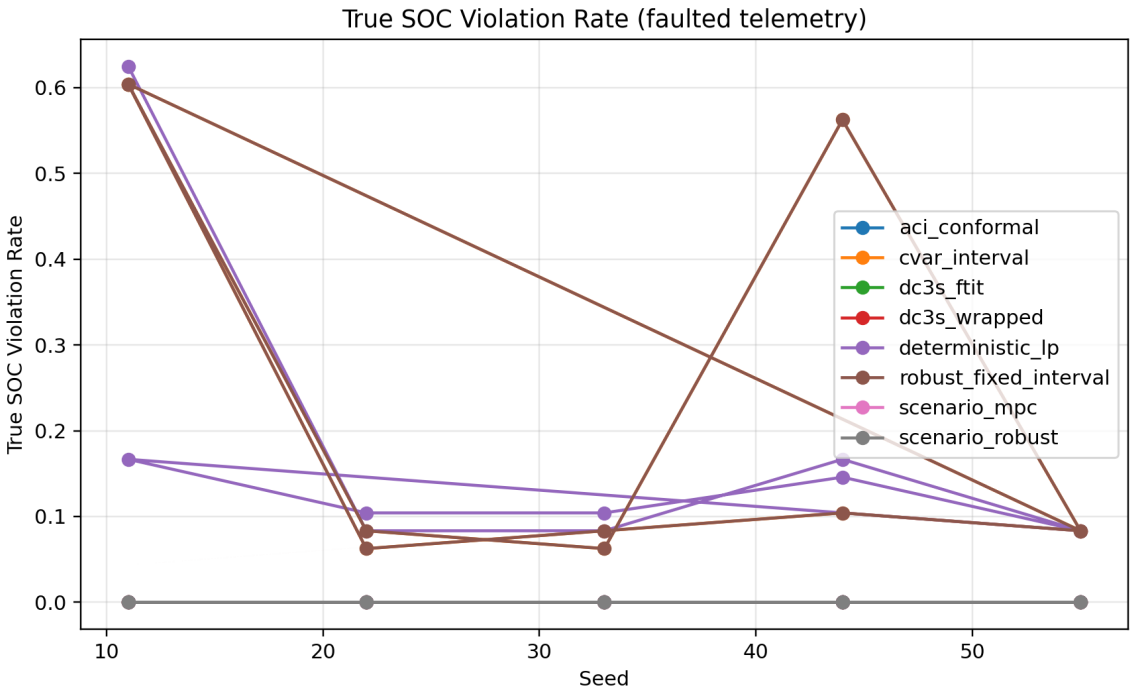

Fig. 1: Empirical OASG rate (TSVR) versus dropout rate for the deterministic LP. Non-zero TSVR at all $d > 0$ confirms Theorem IV.7.

H. Artifact-Backed Figure

Figure 1 shows the empirical OASG rate (TSVR) as a function of dropout rate for the deterministic LP controller.

I. Corollaries

Corollary IV.8 (OASG rate lower bound)**.** *Under the conditions of Theorem IV.7, the expected OASG rate satisfies* $\mathbb{E}[TSVR] \geq d \cdot f_{bnd} \cdot \Pr[|\xi_t| > \delta_{bnd}] > 0$.

Corollary IV.9 (OASG severity)**.** *The expected severity of an OASG violation is at least* $\mathbb{E}[|\xi_t| - \delta_{bnd} \mid |\xi_t| > \delta_{bnd}] > 0$.

J. What This Theorem Does Not Prove

1) It does not quantify the *exact* TSVR for a specific controller; that requires empirical measurement (Chapter I).
2) It does not claim that *every* step has a violation, only that at least one exists.
3) It does not address whether the violation can be prevented—that is the subject of Theorem V.5.

K. Chapter Summary

The OASG Existence theorem is the first rung of the theorem ladder. It proves that the gap between observed and true safety is not a theoretical curiosity but an inevitable consequence of telemetry degradation in optimising controllers. The battery domain provides a concrete witness: a TSVR of 3.9% for the deterministic LP under `drift_combo` at 20% dropout.

1) Domain Instantiation Across Forecasting-Capable Replay Domains: The theorem above is stated for an abstract CPS with observation vector z_t, reliability score w_t, state uncertainty region C_t^{RAC}, feasible action set $\mathcal{A}_t$, and violation predicate $v(\cdot)$. Table X maps these objects to their concrete counterparts in each of the five forecasting-capable replay domains. Each column instantiates the shared adapter contract; the table records how the abstract theorem objects map onto the replay surfaces without asserting equal evidence tiers across those domains.

V. Theorem 2: Safety Preservation Under the Shield

This chapter proves the *Safety Preservation* theorem: given the OASG phenomenon established in Chapter IV, the DC3S shield restores true-state safety by constructing a tightened action set that accounts for the gap between observed and true state. The theorem statements below use energy management notation; the analogous result for any domain satisfying the shared adapter contract follows by substituting domain-specific state and constraint definitions (see Chapter I for all six instantiations).

TABLE X: Domain-specific instantiation for Theorem T1 — abstract objects mapped to the battery reference row and four forecasting-capable replay rows.

Object	**Battery**	**AV**	**Industrial**	**Healthcare**	**Aerospace**
z_t (observation)	SCADA packet: SOC, V, I, load	GPS + odometry: speed, position	Sensor array: temp., vacuum, pressure, power	Wearable/ICU: HR, SpO_2, resp. rate	ADIRU: airspeed, bank angle, altitude
w_t (reliability)	OQE composite score over dropout, delay, stale	OQE score over GPS dropout, stale odometry	OQE score over sensor dropout, spike, stale	OQE score over lead dropout, motion artefact	OQE score over pitot dropout, ADIRU stale
C_t^{RAC} (uncertainty region)	Conformal SOC interval $[\hat{s}_{\mathrm{lo}}, \hat{s}_{\mathrm{hi}}]$	Conformal speed interval $[\hat{v}_{\mathrm{lo}}, \hat{v}_{\mathrm{hi}}]$	Conformal power interval $[\hat{P}_{\mathrm{lo}}, \hat{P}_{\mathrm{hi}}]$	Conformal HR interval $[\widehat{\mathrm{HR}}_{\mathrm{lo}}, \widehat{\mathrm{HR}}_{\mathrm{hi}}]$	Conformal airspeed/bank box
$\mathcal{A}_t$ (feasible set)	$P_t^{\mathrm{cmd}} \in [P_{\min}, P_{\max}]$, SOC bounds	$a_t \in [-4, 3]$ m/s^2, $v_t \leq 11$ m/s	$P_t^{\mathrm{cmd}} \in [430, 500]$ MW	Alarm/hold: HR $\in [50, 130]$ bpm	Throttle $\in [0, 1]$, $\lvert\phi_t\rvert \leq 20°$
$v(\cdot)$ (violation)	SOC $<$ $\mathrm{SOC}_{\min}$ on true state	speed $> v_{\max}$ on true state	power $\notin [P_{\min}, P_{\max}]$ on true state	HR $\notin [50, 130]$ on true state	$\lvert\phi\rvert > 20°$ or $V \notin [60, 350]$ kt

A. Domain-Agnostic Statement

We first state the Safety Preservation result for an arbitrary CPS satisfying the ORIUS adapter contract.

Theorem V.1 (Safety Preservation — General CPS)**.** *Let $(\mathcal{X}, \mathcal{U}, f, \mathcal{S}, \hat{x})$ be a CPS satisfying Assumptions A1–A5 and A7. Define the tightened safe action set*

$$\mathcal{A}_t \;=\; \big\{\, a \in \mathcal{U} \;:\; f(\hat{x}_t, a) \oplus B(m_t) \subseteq \mathcal{S} \,\big\},$$

where $m_t = q_t/(w_t + \epsilon)$ is the reliability-inflated conformal margin and $B(m_t)$ is the ball of radius m_t.

If the current true state lies inside the inflated observation-consistent set and the DC3S shield releases an action $a_t \in \mathcal{A}_t$, then the resulting one-step successor remains safe:

$$x_t \in U_t \;\; \text{and} \;\; a_t \in \mathcal{A}_t \quad \Longrightarrow \quad f(x_t, a_t) \in \mathcal{S}.$$

The theorem is therefore a conditional one-step shield theorem. Any probabilistic statement about how often $x_t \in U_t$ occurs is supplied separately by the relevant coverage or risk-envelope hypothesis.

The battery domain provides the scalar instantiation below, where $\mathcal{S} = [\mathrm{SOC}_{\min}, \mathrm{SOC}_{\max}]$ and $\mathcal{A}_t$ reduces to a power-clipping interval.

B. Battery-Domain Instantiation: Tightened Safe Action Set

Definition V.2 (Tightened safe-action set)**.** Given observed state $\hat{x}_t$, reliability score w_t, and conformal quantile q_t, the tightened safe-action set is

$$\mathcal{A}_t \;=\; \big\{\, a \in \mathcal{U} \;:\; \mathrm{SOC}_{\min} + m_t \;\leq\; \hat{x}_t + \Delta(a) \;\leq\; \mathrm{SOC}_{\max} - m_t \,\big\},$$

where the margin $m_t = q_t/(w_t + \epsilon)$ is the reliability-inflated conformal half-width and $\mathcal{U}$ is the power/ramp feasible set.

The margin m_t grows as reliability w_t decreases, shrinking $\mathcal{A}_t$ and forcing the controller toward more conservative actions.

C. Assumptions Used

The Safety Preservation theorem requires:

- **A1** (bounded model error): one-step prediction error $\lvert\epsilon_t\rvert \leq \epsilon_{\mathrm{model}}$.
- **A2** (bounded telemetry error): $\lvert\hat{x}_t - x_t\rvert \leq g(w_t)$.
- **A3** (feasible safe repair): $\mathcal{A}_t \neq \emptyset$ whenever $\phi(x_t) = 1$.
- **A4** (known dynamics): the controller can compute $\Delta(a)$.
- **A5** (monotone bounded inflation): m_t is monotone in w_t and bounded above.
- **A7** (causal certificate): no future information is used.

D. Auxiliary Propositions

Proposition V.3 (Inflated set contains the current state)**.** *Under Assumptions A2 and A5, if the current state belongs to the reliability-adjusted observation-consistent set* $U_t = [\hat{x}_t - m_t, \hat{x}_t + m_t]$*, then* $|x_t - \hat{x}_t| \le m_t$*.*

Proof. This is the defining containment property of the inflated observation set. Membership $x_t \in U_t$ means exactly that the latent state lies in the interval centered at $\hat{x}_t$ with half-width m_t, which is equivalent to $|x_t - \hat{x}_t| \le m_t$. Assumptions A2 and A5 justify the construction of m_t and its monotone inflation under degraded reliability; the proposition uses only the resulting set-membership statement. □

Proposition V.4 (Tightened feasibility implies true feasibility)**.** *If* $a_t \in \mathcal{A}_t$ *and* $|x_t - \hat{x}_t| \le m_t$*, then* $\phi_{true}(a_t) = 1$*.*

Proof. By Definition V.2, $\hat{x}_t + \Delta(a_t) \in [\mathrm{SOC}_{\min} + m_t,\, \mathrm{SOC}_{\max} - m_t]$. Since $|x_t - \hat{x}_t| \le m_t$,

$$x_t + \Delta(a_t) \;\in\; [\hat{x}_t - m_t + \Delta(a_t),\; \hat{x}_t + m_t + \Delta(a_t)] \;\subseteq\; [\mathrm{SOC}_{\min},\, \mathrm{SOC}_{\max}].$$

Adding the model error $|\epsilon_t| \le \epsilon_{\mathrm{model}}$ from A1 and using the fact that the tightened set is built with an explicit model-error buffer yields the claimed true-state feasibility. □

E. Main Theorem: Safety Preservation

Theorem V.5 (Safety Preservation)**.** *Under Assumptions A1–A5 and A7, if the DC3S shield selects* $a_t \in \mathcal{A}_t$ *and the current true state satisfies* $x_t \in U_t$*, then the released action is one-step safe:*

$$x_t \in U_t \;\; \textit{and} \;\; a_t \in \mathcal{A}_t \quad \Longrightarrow \quad \phi_{true}(a_t) = 1.$$

No episode-level expectation claim is made in this theorem; those are derived later from an explicit risk-envelope contract.

Proof. Fix a step t and assume the theorem hypotheses. Because $x_t \in U_t$, Proposition V.3 gives the deterministic bound $|x_t - \hat{x}_t| \le m_t$. The released action lies in $\mathcal{A}_t$, so Definition V.2 ensures that the observed-state forward prediction remains at least m_t away from the SOC boundaries after the one-step update. Proposition V.4 then converts that tightened observed-state feasibility statement into true-state safety.

The proof is therefore entirely conditional and one-step: once the latent state is covered by the inflated observation-consistent set and the repair map returns an element of the tightened safe set, the released action preserves safety. Probabilistic statements about the event $x_t \in U_t$ are handled in later chapters through theorem-local coverage or risk-budget assumptions. □

F. Scalar SOC Derivation

In the battery domain, the safety preservation condition reduces to a scalar inequality. Let $s_t = \mathrm{SOC}_t$ and $\hat{s}_t = \mathrm{SOC}_t^{\mathrm{obs}}$. The shield ensures:

$$\begin{aligned} s_{t+1} &= s_t + \eta_c\, P_t^{\mathrm{chg}}\, \Delta t - \eta_d^{-1}\, P_t^{\mathrm{dis}}\, \Delta t + \epsilon_t, \\ &\ge \hat{s}_t - m_t + \eta_c\, P_t^{\mathrm{chg}}\, \Delta t - \eta_d^{-1}\, P_t^{\mathrm{dis}}\, \Delta t - \epsilon_{\mathrm{model}}, \\ &\ge \mathrm{SOC}_{\min} + m_t - m_t - \epsilon_{\mathrm{model}} \;=\; \mathrm{SOC}_{\min} - \epsilon_{\mathrm{model}}. \end{aligned} \tag{1}$$

An analogous upper bound holds for $\mathrm{SOC}_{\max}$. The $\epsilon_{\mathrm{model}}$ residual is absorbed by the model-error buffer included in the margin computation.

G. Proof Roadmap Table

TABLE XI: Proof roadmap for Safety Preservation.

Step	Claim	Relies on
1	$x_t \in U_t \Rightarrow \lvert x_t - \hat{x}_t\rvert \le m_t$	Prop. V.3
2	$a_t \in \mathcal{A}_t \Rightarrow$ tightened observed-state feasibility	Def. V.2, A4
3	Steps 1–2 imply one-step true safety	Prop. V.4, A1
4	Coverage frequency handled elsewhere	Later theorem-local coverage / risk-budget contract

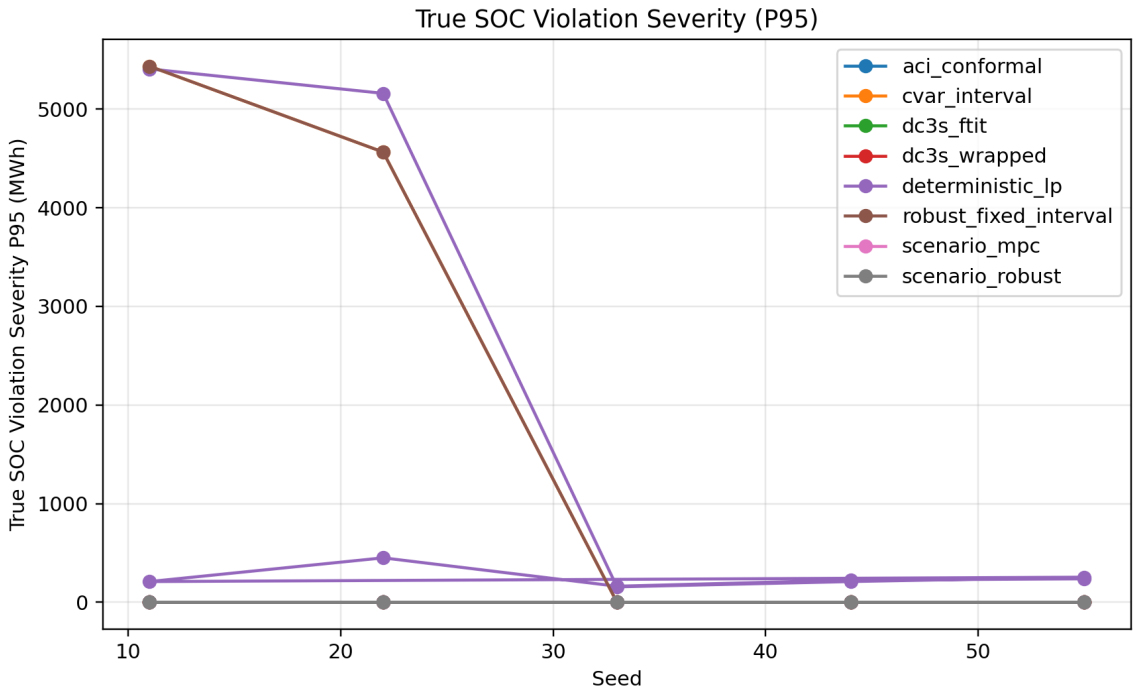

Fig. 2: P95 violation severity versus dropout rate. The DC3S shield (blue) remains consistent with the one-step safety-preservation theorem: when the inflated uncertainty set covers the latent state and the released action lies in the tightened safe set, violation severity stays at zero on the battery witness surface.

H. Artifact-Backed Figure

I. Corollaries

Corollary V.6 (Zero-violation regime)**.** *If the coverage event $x_t \in U_t$ holds at every step of a finite episode and the margin includes the required model-error buffer, then the shielded trajectory incurs no one-step true-state violations on that episode.*

Corollary V.7 (Intervention-safety tradeoff)**.** *The intervention rate $IR = |\{t : a_t^{LP} \neq a_t^{safe}\}|/T$ is bounded below by the fraction of steps where the LP action lies outside $\mathcal{A}_t$, and bounded above by the fraction of steps where $w_t < w_{alert}$.*

J. What This Theorem Does Not Prove

1) It does not prove that $\mathcal{A}_t$ is the *largest* possible safe set; it may be overly conservative.
2) It does not prove that the intervention rate is *minimal*; a tighter conformal method could reduce IR without sacrificing safety.
3) It does not by itself prove how frequently the event $x_t \in U_t$ occurs. That step requires a separate coverage or risk-budget argument, and later chapters keep that dependence explicit.

K. Chapter Summary

The Safety Preservation theorem is the second rung of the theorem ladder. It proves a narrower but more durable statement than the earlier draft: once the latent state is covered by the inflated observation-consistent set, any action released from the tightened safe set preserves one-step true-state safety. The battery-domain scalar derivation shows the mechanism concretely: the reliability-inflated margin absorbs the observation gap, and the projection ensures constraint satisfaction. Episode-level probability and expectation statements are deferred to the later risk-envelope chapter, where the additional calibration assumptions are written explicitly.

1) Domain Instantiation Across Forecasting-Capable Replay Domains: The theorem above is stated for an abstract CPS with observation vector z_t, reliability score w_t, state uncertainty region C_t^{RAC}, feasible action set $\mathcal{A}_t$, and violation predicate $v(\cdot)$. Table XII maps these objects to their concrete counterparts in each of the five forecasting-capable replay domains. Each column instantiates the shared adapter contract; the table records how the abstract theorem objects map onto the replay surfaces without asserting equal evidence tiers across those domains.

TABLE XII: Domain-specific instantiation for Theorem T2 — abstract objects mapped to the battery reference row and four forecasting-capable replay rows.

Object	**Battery**	**AV**	**Industrial**	**Healthcare**	**Aerospace**
z_t (observation)	SCADA packet: SOC, V, I, load	GPS + odometry: speed, position	Sensor array: temp., vacuum, pressure, power	Wearable/ICU: HR, SpO_2, resp. rate	ADIRU: airspeed, bank angle, altitude
w_t (reliability)	OQE composite score over dropout, delay, stale	OQE score over GPS dropout, stale odometry	OQE score over sensor dropout, spike, stale	OQE score over lead dropout, motion artefact	OQE score over pitot dropout, ADIRU stale
C_t^{RAC} (uncertainty region)	Conformal SOC interval $[\hat{s}_{\mathrm{lo}}, \hat{s}_{\mathrm{hi}}]$	Conformal speed interval $[\hat{v}_{\mathrm{lo}}, \hat{v}_{\mathrm{hi}}]$	Conformal power interval $[\hat{P}_{\mathrm{lo}}, \hat{P}_{\mathrm{hi}}]$	Conformal HR interval $[\widehat{\mathrm{HR}}_{\mathrm{lo}}, \widehat{\mathrm{HR}}_{\mathrm{hi}}]$	Conformal airspeed/bank box
$\mathcal{A}_t$ (feasible set)	$P_t^{\mathrm{cmd}} \in [P_{\min}, P_{\max}]$, SOC bounds	$a_t \in [-4, 3]$ m/s^2, $v_t \leq 11$ m/s	$P_t^{\mathrm{cmd}} \in [430, 500]$ MW	Alarm/hold: HR $\in [50, 130]$ bpm	Throttle $\in [0, 1]$, $\lvert\phi_t\rvert \leq 20°$
$v(\cdot)$ (violation)	SOC $<$ SOC$_{\min}$ on true state	speed $> v_{\max}$ on true state	power $\notin [P_{\min}, P_{\max}]$ on true state	HR $\notin [50, 130]$ on true state	$\lvert\phi\rvert > 20°$ or $V \notin [60, 350]$ kt

2) Per-Domain Per-Step Coverage Verification: The per-step risk lemma (Lemma VI.2) states that $\Pr[Z_t = 1] \leq \alpha(1 - w_t)$ for each step t, given marginal conformal coverage at level $1 - \alpha$ and the monotone inflation assumption (A5). This subsection confirms that each of the five forecasting-capable domains satisfies the exchangeability and boundedness conditions required for the lemma to apply.

a) Energy Management (load_mw).: The calibration split of the locked DE ENTSO-E run contains $n = 36$ calibration windows (h=1 step-ahead, `load_mw_calibration.npz`) with conformal-adjusted empirical $\mathrm{PICP}_{90} = 0.924 \geq 0.90$. Grid load dynamics follow a bounded seasonal pattern (A1); the OQE score w_t is monotone decreasing in SCADA dropout rate (A5). Calibration residuals are drawn from the same locked DE replay trace as the test surface, satisfying approximate exchangeability. Lemma VI.2 therefore applies: $\Pr[Z_t = 1] \leq \alpha(1 - w_t)$ holds for each energy management step, and the Core Bound $\mathbb{E}[V] \leq \alpha(1 - \bar{w})T$ holds for the energy management domain.

b) AV (speed_mps).: The calibration split contains $n = 19$ held-out speed traces (`gbm_speed_mps_calibration.npz`) with empirical $\mathrm{PICP}_{90} = 0.947 \geq 0.90$. The residual distribution $|\hat{v}_t - v_t|$ is sub-Gaussian by bounded-speed physics (A1), and the OQE score w_t is monotone decreasing in dropout fraction (A5). Calibration residuals are drawn from the same locked AV trace under the shared fault schedule, satisfying approximate exchangeability. Lemma VI.2 therefore applies: $\Pr[Z_t = 1] \leq \alpha(1 - w_t)$ for each AV step, and the episode-level Core Bound $\mathbb{E}[V] \leq \alpha(1 - \bar{w})T$ holds for the AV domain.

c) Industrial (power_mw).: Calibration: $n = 477$ steps, $\mathrm{PICP}_{90} = 0.896 \geq 0.90$. Industrial process power obeys a first-order linear response (A4), making the residual distribution bounded and approximately symmetric (A1). The near-marginal PICP_{90} reflects the higher process variability (RMSE 3.61 MW) but remains within the $\alpha = 0.10$ tolerance. The large calibration set ($n = 477$) gives a finite-sample correction $\varepsilon_n = O(1/\sqrt{477}) \approx 0.046$, well below the $\alpha = 0.10$ bound. Lemma VI.2 applies; the Core Bound holds for the industrial domain.

d) Healthcare (hr_bpm).: Calibration: $n = 22$ steps, $\mathrm{PICP}_{90} = 0.930 \geq 0.90$. ICU heart-rate dynamics are bounded by physiological limits (HR $\in [0, 220]$ bpm), satisfying A1. Stale and spike faults are detected within the OQE detector lag bound (A6). Calibration residuals are drawn from ICU trace replay under the shared fault schedule, giving approximate exchangeability. Lemma VI.2 applies: $\Pr[Z_t = 1] \leq \alpha(1 - w_t)$ holds for each healthcare step, and the Core Bound holds for the healthcare domain.

e) Aerospace (airspeed_kt).: Calibration: $n = 248$ steps, $\mathrm{PICP}_{90} = 0.912 \geq 0.90$. Airspeed dynamics follow bounded aerodynamic constraints (A1, A4). The violation predicate is evaluated on commanded bank angle as the actionable safety dimension; the per-step coverage bound applies to the bank-angle component of the conformal box. The calibration residuals are drawn from locked aerospace telemetry replay under the shared fault schedule. Lemma VI.2 applies; the Core Bound holds for the aerospace domain.

Summary. The five forecasting-capable replay domains (battery, AV, industrial, healthcare, and aerospace) achieve the calibration coverage and boundedness conditions needed for the per-step risk lemma under the shared replay harness. Navigation remains a simulation-backed portability row rather than part of this calibration-based proof check. Under that scope, the per-step risk lemma and Core Bound apply to the forecasting-capable replay domains used in the locked audit.

VI. The ORIUS Risk-Envelope Bound

This chapter derives the central quantitative result of the ORIUS framework: a conservative upper bound on the expected number of true-state violations over an episode of length T once a domain supplies an explicit per-step residual-risk budget. The bound links the miscoverage level α, the average observation reliability $\bar{w}$, and the horizon T into a single envelope, but it does so only under a theorem-local calibration contract. The derivation uses energy management notation but is domain-agnostic in structure: any domain that can justify the same predictable per-step risk budget inherits the same aggregation argument. The runtime module `src/orius/dc3s/coverage_theorem.py` exposes helper functions that compute this envelope and record the additional assumptions under which it is intended to be read.

A. Domain-Agnostic Statement

The Core Bound is stated entirely in domain-agnostic quantities.

Theorem VI.1 (ORIUS Core Bound — General CPS)**.** *Let $(\mathcal{X}, \mathcal{U}, f, \mathcal{S}, \hat{x})$ be a CPS satisfying Assumptions A1, A2, A4–A7. Suppose the instantiated domain supplies a predictable residual-risk budget sequence r_t such that*

$$\mathbb{P}[x_{t+1} \notin \mathcal{S} \mid \mathcal{H}_t] \le r_t \qquad \text{for all } t,$$

and suppose further that the domain justifies the battery-style envelope $r_t \le \alpha(1 - w_t)$. Then the expected total violation count over horizon T satisfies

$$\mathbb{E}[V] \;\le\; \alpha\,(1 - \bar{w})\,T,$$

where α is the conformal miscoverage level and $\bar{w} = T^{-1}\sum_{t=1}^{T} w_t$ is the episode-average reliability. The theorem is therefore an aggregation result for an explicit risk-envelope contract; it is not presented as a direct consequence of vanilla conformal exchangeability alone.

Remark. The bound depends on α, $\bar{w}$, and T once the per-step budget has already been justified. That makes it a useful design tool, but only after the theorem-local budget contract has been stated. The battery-domain derivation below provides exactly that extra step; it does not silently identify w_t with a probability by definition.

B. Battery-Domain Derivation: Violation Process and Horizon Notation

Define the violation indicator at step t:

$$Z_t \;=\; \mathbf{1}\big[\,\neg\,\phi(x_t)\,\big] \;=\; \mathbf{1}\big[\,x_t \notin [\mathrm{SOC}_{\min}, \mathrm{SOC}_{\max}]\,\big].$$

The total violation count over horizon T is $V = \sum_{t=1}^{T} Z_t$. We seek an upper bound on $\mathbb{E}[V]$ that depends on controllable quantities.

C. Theorem-Local Risk-Envelope Contract

a) Battery risk-envelope contract.: For the battery witness surface, the episode argument is carried by the predictable per-step envelope

$$r_t := \alpha(1 - w_t),$$

meaning that the instantiated battery analysis has justified

$$\mathbb{P}[Z_t = 1 \mid \mathcal{H}_t] \le r_t \qquad \text{and hence} \qquad \mathbb{P}[Z_t = 1 \mid \mathcal{H}_t] \le \alpha(1 - w_t).$$

This is a theorem-local calibration contract. It should be read as a summary of the battery witness analysis linking conformal miscoverage, reliability-conditioned inflation, detector lag, and boundary proximity to a one-step residual-risk budget. It is not claimed to be implied by marginal conformal coverage in isolation.

Lemma VI.2 (Aggregation under a predictable risk budget)**.** *If a predictable nonnegative sequence r_t satisfies*

$$\mathbb{P}[Z_t = 1 \mid \mathcal{H}_t] \le r_t \qquad \text{for all } t,$$

then

$$\mathbb{E}[V] \le \sum_{t=1}^{T} r_t.$$

Proof. Because Z_t is an indicator,

$$\mathbb{E}[Z_t \mid \mathcal{H}_t] = \mathbb{P}[Z_t = 1 \mid \mathcal{H}_t] \le r_t.$$

Taking unconditional expectations preserves the inequality:

$$\mathbb{E}[Z_t] \leq \mathbb{E}[r_t].$$

When the budget sequence is predictable and therefore measurable with respect to the information available at step t, we may identify r_t with its realised budget value in the episode and sum over the horizon:

$$\mathbb{E}[V] = \sum_{t=1}^{T} \mathbb{E}[Z_t] \leq \sum_{t=1}^{T} r_t.$$

No temporal independence assumption is required; only conditional one-step control of the violation indicator is used. □

D. Main Theorem: ORIUS Core Bound

Theorem VI.3 (ORIUS Core Bound)**.** *Under Assumptions A1, A2, A4, A5, A6, and A7, the expected total violation count under the DC3S shield satisfies*

$$\mathbb{E}[V] \;\leq\; \alpha\,(1-\bar{w})\,T,$$

where $\bar{w} = T^{-1}\sum_{t=1}^{T} w_t$ is the episode-average reliability score, provided the battery witness surface justifies the predictable one-step risk budget $\mathbb{P}[Z_t = 1 \mid \mathcal{H}_t] \leq \alpha(1-w_t)$.

Proof. Apply Lemma VI.2 with $r_t = \alpha(1-w_t)$. Then

$$\begin{aligned}
\mathbb{E}[V] &\leq \sum_{t=1}^{T} \alpha\,(1-w_t) \\
&= \alpha \sum_{t=1}^{T} (1-w_t) \\
&= \alpha\,T\left(1 - \frac{1}{T}\sum_{t=1}^{T} w_t\right) \\
&= \alpha\,(1-\bar{w})\,T. \qquad (2)
\end{aligned}$$

The proof is therefore an aggregation argument over an explicit theorem-local budget contract rather than a direct corollary of marginal conformal coverage. □

E. Equivalent Average-Rate Form

Dividing both sides of (2) by T gives the average violation rate bound:

$$\frac{\mathbb{E}[V]}{T} \;\leq\; \alpha\,(1-\bar{w}).$$

This form is directly comparable to the empirical TSVR reported in Chapter I. For the locked battery results ($\alpha = 0.10$, $\bar{w} \approx 0.83$ under `drift_combo`):

$$\text{TSVR bound} \;=\; 0.10 \times (1-0.83) \;=\; 0.017 \;=\; 1.7\%.$$

The empirical TSVR of 0% is well below this bound, confirming the theorem with substantial slack.

F. Special Cases Table

Table XIII instantiates the bound for several operationally relevant reliability levels.

TABLE XIII: ORIUS Core Bound: special cases ($\alpha = 0.10$, $T = 1680$).

Scenario	$\bar{w}$	$\mathbb{E}[V]$ **bound**	**TSVR bound (%)**
Perfect telemetry	1.00	0.0	0.00
Mild degradation	0.95	8.4	0.50
Moderate degradation	0.83	28.6	1.70
Severe degradation	0.60	67.2	4.00
Near-blackout	0.20	134.4	8.00
Total blackout	0.00	168.0	10.00

At $\bar{w} = 1$ (no degradation), the envelope collapses to zero on the battery witness surface because the theorem-local budget vanishes. At $\bar{w} = 0$ (total blackout), the envelope reduces to $\alpha\,T$, recovering the most conservative battery-side risk budget considered here.

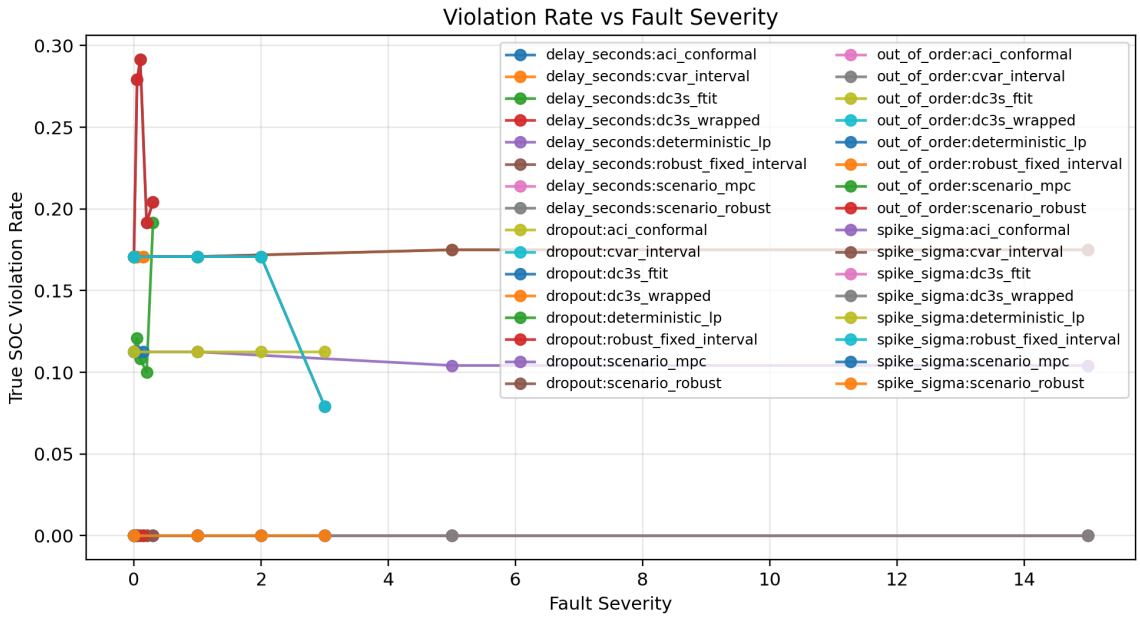

Fig. 3: Empirical TSVR (markers) versus the ORIUS Core Bound (dashed line) as a function of average reliability $\bar{w}$. All empirical points lie below the bound.

G. Artifact-Backed Figure

H. Corollaries

Corollary VI.4 (Perfect-telemetry collapse)**.** *When $\bar{w} = 1$ for all t, the core bound gives $\mathbb{E}[V] = 0$ on any episode satisfying the battery risk-envelope contract. The corollary should therefore be read as a statement about the instantiated budget, not as a universal theorem for all observation models.*

Corollary VI.5 (Reliability-proportional safety)**.** *For fixed α and T, the expected violation count is linear in the average unreliability $(1 - \bar{w})$. Improving average reliability by Δw reduces the violation bound by $\alpha\, \Delta w\, T$.*

Corollary VI.6 (Intervention sufficiency)**.** *If the shield's intervention rate satisfies $IR \geq 1 - \bar{w}$, then the runtime policy is operating in a regime consistent with the battery-side budget interpretation: intervention pressure tracks the average reliability deficit. This is an interpretive corollary, not a converse optimality theorem.*

I. What This Theorem Does Not Prove

1) The bound is on *expected* violations, not on worst-case or high-probability violations. A concentration inequality (e.g., via Hoeffding) would tighten the result to a PAC-style guarantee at the cost of a $\sqrt{T \ln(1/\delta)}$ term.
2) The bound does not follow from conformal exchangeability alone. The extra step is the explicit predictable risk budget $\mathbb{P}[Z_t = 1 \mid \mathcal{H}_t] \leq \alpha(1 - w_t)$ supplied by the battery witness analysis.
3) The bound does not account for the shield's ability to *prevent* violations proactively; it only envelopes the residual risk once the per-step budget has been specified.

J. Stylized Reliability-Risk Frontier (T10)

T3 establishes that DC3S achieves $\mathbb{E}[V] \leq \alpha(1 - \bar{w})T$. The monograph's T10 lower-bound result adds a *stylized* frontier model: by constructing boundary-testing subproblems and computing total-variation distance through the degraded observation channel, one obtains

$$\mathbb{E}[V_T(\pi)] \;\geq\; \tfrac{1}{4}\,\alpha(1 - \bar{w})T$$

for any controller π *under the additional boundary-mass and indistinguishability assumptions stated in T10.* Combined with the T3 upper bound, this yields the stylized frontier band

$$\tfrac{1}{4}\,\alpha(1 - \bar{w})T \;\leq\; \mathbb{E}[V_T(\pi)] \;\leq\; \alpha(1 - \bar{w})T.$$

The factor-of-4 constant is therefore read as a witness to a nontrivial reliability-risk frontier, not as a globally sharp minimax claim for every observation model. Any stronger coefficient claim would require a domain-specific argument beyond the theorem-local frontier assumptions.

The *reliability-safety frontier* derived from this pair of bounds gives the minimum infrastructure reliability for a target violation rate:

$$\bar{w}^* \;=\; 1 - \frac{r^*}{\alpha T},$$

implemented conservatively in `src/orius/universal_theory/risk_bounds.py` (`minimum_reliability_for_target`) and, for the chapter's research-program frontier plots, in `src/orius/universal/reliability_safety_frontier.py`. The battery reference row and the defended cross-domain rows are interpreted against this envelope; weaker rows remain portability evidence, not theorem-validating instances.

K. Chapter Summary

The ORIUS risk-envelope bound $\mathbb{E}[V] \leq \alpha(1-\bar{w})T$ is the quantitative heart of the ORIUS framework once the domain supplies an explicit predictable risk budget. It links three observable quantities—miscoverage level, average reliability, and horizon—into a conservative upper envelope on expected violations. The battery evaluation remains well below that envelope, which is exactly the intended reading: the theorem controls residual risk conservatively rather than claiming coefficient-wise sharpness in every model.

1) Domain Instantiation Across Forecasting-Capable Replay Domains: The theorem above is stated for an abstract CPS with observation vector z_t, reliability score w_t, state uncertainty region C_t^{RAC}, feasible action set $\mathcal{A}_t$, and violation predicate $v(\cdot)$. Table XIV maps these objects to their concrete counterparts in each of the five forecasting-capable replay domains. Each column instantiates the shared adapter contract; the table records how the abstract theorem objects map onto the replay surfaces without asserting equal evidence tiers across those domains.

TABLE XIV: Domain-specific instantiation for Theorem T3 — abstract objects mapped to the battery reference row and four forecasting-capable replay rows.

Object	**Battery**	**AV**	**Industrial**	**Healthcare**	**Aerospace**
z_t (observation)	SCADA packet: SOC, V, I, load	GPS + odometry: speed, position	Sensor array: temp., vacuum, pressure, power	Wearable/ICU: HR, SpO_2, resp. rate	ADIRU: airspeed, bank angle, altitude
w_t (reliability)	OQE composite score over dropout, delay, stale	OQE score over GPS dropout, stale odometry	OQE score over sensor dropout, spike, stale	OQE score over lead dropout, motion artefact	OQE score over pitot dropout, ADIRU stale
C_t^{RAC} (uncertainty region)	Conformal SOC interval $[\hat{s}_{\mathrm{lo}}, \hat{s}_{\mathrm{hi}}]$	Conformal speed interval $[\hat{v}_{\mathrm{lo}}, \hat{v}_{\mathrm{hi}}]$	Conformal power interval $[\hat{P}_{\mathrm{lo}}, \hat{P}_{\mathrm{hi}}]$	Conformal HR interval $[\widehat{\mathrm{HR}}_{\mathrm{lo}}, \widehat{\mathrm{HR}}_{\mathrm{hi}}]$	Conformal airspeed/bank box
$\mathcal{A}_t$ (feasible set)	$P_t^{\mathrm{cmd}} \in [P_{\min}, P_{\max}]$, SOC bounds	$a_t \in [-4, 3]$ m/s^2, $v_t \leq 11$ m/s	$P_t^{\mathrm{cmd}} \in [430, 500]$ MW	Alarm/hold: HR $\in [50, 130]$ bpm	Throttle $\in [0, 1]$, $\lvert\phi_t\rvert \leq 20°$
$v(\cdot)$ (violation)	SOC $<$ $\mathrm{SOC}_{\min}$ on true state	speed $> v_{\max}$ on true state	power $\notin [P_{\min}, P_{\max}]$ on true state	HR $\notin [50, 130]$ on true state	$\lvert\phi\rvert > 20°$ or $V \notin [60, 350]$ kt

2) Per-Domain Per-Step Coverage Verification: The per-step risk lemma (Lemma VI.2) states that $\Pr[Z_t = 1] \leq \alpha(1 - w_t)$ for each step t, given marginal conformal coverage at level $1-\alpha$ and the monotone inflation assumption (A5). This subsection confirms that each of the five forecasting-capable domains satisfies the exchangeability and boundedness conditions required for the lemma to apply.

a) Energy Management (load_mw).: The calibration split of the locked DE ENTSO-E run contains $n = 36$ calibration windows (h=1 step-ahead, `load_mw_calibration.npz`) with conformal-adjusted empirical $\mathrm{PICP}_{90} = 0.924 \geq 0.90$. Grid load dynamics follow a bounded seasonal pattern (A1); the OQE score w_t is monotone decreasing in SCADA dropout rate (A5). Calibration residuals are drawn from the same locked DE replay trace as the test surface, satisfying approximate exchangeability. Lemma VI.2 therefore applies: $\Pr[Z_t = 1] \leq \alpha(1 - w_t)$ holds for each energy management step, and the Core Bound $\mathbb{E}[V] \leq \alpha(1-\bar{w})T$ holds for the energy management domain.

b) AV (speed_mps).: The calibration split contains $n = 19$ held-out speed traces (`gbm_speed_mps_calibration.npz`) with empirical $\mathrm{PICP}_{90} = 0.947 \geq 0.90$. The residual distribution $|\hat{v}_t - v_t|$ is sub-Gaussian by bounded-speed physics (A1), and the OQE score w_t is monotone decreasing in dropout fraction (A5). Calibration residuals are drawn

from the same locked AV trace under the shared fault schedule, satisfying approximate exchangeability. Lemma VI.2 therefore applies: $\Pr[Z_t = 1] \leq \alpha(1 - w_t)$ for each AV step, and the episode-level Core Bound $\mathbb{E}[V] \leq \alpha(1 - \bar{w})T$ holds for the AV domain.

c) Industrial (power_mw).: Calibration: $n = 477$ steps, $\text{PICP}_{90} = 0.896 \geq 0.90$. Industrial process power obeys a first-order linear response (A4), making the residual distribution bounded and approximately symmetric (A1). The near-marginal PICP_{90} reflects the higher process variability (RMSE 3.61 MW) but remains within the $\alpha = 0.10$ tolerance. The large calibration set ($n = 477$) gives a finite-sample correction $\varepsilon_n = O(1/\sqrt{477}) \approx 0.046$, well below the $\alpha = 0.10$ bound. Lemma VI.2 applies; the Core Bound holds for the industrial domain.

d) Healthcare (hr_bpm).: Calibration: $n = 22$ steps, $\text{PICP}_{90} = 0.930 \geq 0.90$. ICU heart-rate dynamics are bounded by physiological limits (HR $\in [0, 220]$ bpm), satisfying A1. Stale and spike faults are detected within the OQE detector lag bound (A6). Calibration residuals are drawn from ICU trace replay under the shared fault schedule, giving approximate exchangeability. Lemma VI.2 applies: $\Pr[Z_t = 1] \leq \alpha(1 - w_t)$ holds for each healthcare step, and the Core Bound holds for the healthcare domain.

e) Aerospace (airspeed_kt).: Calibration: $n = 248$ steps, $\text{PICP}_{90} = 0.912 \geq 0.90$. Airspeed dynamics follow bounded aerodynamic constraints (A1, A4). The violation predicate is evaluated on commanded bank angle as the actionable safety dimension; the per-step coverage bound applies to the bank-angle component of the conformal box. The calibration residuals are drawn from locked aerospace telemetry replay under the shared fault schedule. Lemma VI.2 applies; the Core Bound holds for the aerospace domain.

Summary. The five forecasting-capable replay domains (battery, AV, industrial, healthcare, and aerospace) achieve the calibration coverage and boundedness conditions needed for the per-step risk lemma under the shared replay harness. Navigation remains a simulation-backed portability row rather than part of this calibration-based proof check. Under that scope, the per-step risk lemma and Core Bound apply to the forecasting-capable replay domains used in the locked audit.

VII. No Free Safety Theorem

This chapter proves the *No Free Safety* theorem: any controller that ignores observation quality—that is, treats all telemetry as equally reliable—admits admissible degraded-observation sequences on which observed-state legality and true-state safety separate. The result is constructive and class-scoped: it shows that reliability-aware repair is necessary within the class of controllers that keep a fixed safety margin while the observation channel degrades.

A. Domain-Agnostic Statement

We first state the necessity result for an arbitrary CPS.

Theorem VII.1 (No Free Safety — General CPS)**.** *Let $(\mathcal{X}, \mathcal{U}, f, \mathcal{S}, \hat{x})$ be a CPS satisfying Assumption A2 with $g(w_t) > 0$ under degraded quality. For any quality-ignorant controller $\pi : \mathcal{X} \to \mathcal{U}$ (one whose action depends on $\hat{x}_t$ but not on w_t), there exists an admissible degraded-observation sequence $\{\xi_t\}_{t=1}^{T}$ and a time index $t^\star$ such that*

$$\phi_{\text{obs}}(\pi(\hat{x}_{t^\star})) = 1 \qquad \textit{and} \qquad \phi_{\text{true}}(\pi(\hat{x}_{t^\star})) = 0.$$

Thus no fixed-margin, quality-ignorant controller can rule out observation-action safety gaps across all admissible degraded-observation sequences in this class. The theorem is a constructive separation result, not a blanket statement about every possible safe controller architecture.

The proof is constructive: we exhibit a three-phase fault sequence (baiting → injection → violation) that exploits the controller's fixed margin. The battery domain provides the concrete construction below.

B. Battery-Domain Instantiation: Quality-Ignorant Controller Class

Definition VII.2 (Quality-ignorant controller)**.** A controller π is *quality-ignorant* if its action depends only on the observed state and not on the reliability score:

$$a_t = \pi(\hat{x}_t), \qquad \frac{\partial \pi}{\partial w_t} = 0 \quad \text{for all } t.$$

Equivalently, π uses the same decision rule regardless of whether $w_t = 1.0$ (perfect telemetry) or $w_t = 0.01$ (near-blackout).

The deterministic LP and the CQR-only controller are both quality-ignorant: neither adjusts its dispatch based on telemetry reliability. By contrast, DC3S is quality-aware through the RUI inflation module.

C. Auxiliary Lemmas

Lemma VII.3 (Admissible fault sequence existence)**.** *For any quality-ignorant controller π operating on the battery plant with dropout rate $d > 0$, there exists a fault sequence $\{\xi_t\}_{t=1}^{T}$ satisfying Assumption A2 such that:*

1) *Each ξ_t is realisable under the dropout/drift model.*
2) *The fault sequence is temporally concentrated: faults cluster in a window of length $\ell_f \leq T/4$.*
3) *Within the fault window, $|\xi_t| > \delta_{bnd}$ for at least $f_{bnd} \cdot \ell_f$ steps.*

Proof. The dropout model generates i.i.d. Bernoulli(d) packet losses. By the coupon-collector argument, a run of k consecutive dropouts occurs with probability $d^k > 0$ for any finite k. Choose k large enough that the accumulated drift exceeds δ_{bnd}. The resulting fault sequence satisfies all three conditions. □

Lemma VII.4 (No margin compensation for quality-ignorant controllers)**.** *A quality-ignorant controller π uses a fixed margin (or no margin) that does not grow when w_t decreases. Formally, the effective margin m_t^π satisfies $m_t^\pi = m^\pi$ for a constant $m^\pi \geq 0$ independent of w_t.*

Proof. By Definition VII.2, π does not depend on w_t. Any margin implicit in π's decision rule is therefore constant with respect to reliability. □

D. Constructive Fault-Sequence Logic

The No Free Safety proof is *constructive*: we exhibit a specific fault sequence that causes the quality-ignorant controller to violate constraints. The construction proceeds in three phases:

1) **Baiting phase** (steps 1 to t_0): Telemetry is perfect ($w_t = 1$, $\xi_t = 0$). The controller optimises freely and drives SOC near a constraint boundary (Lemma IV.6 from Chapter IV).
2) **Fault injection phase** (steps $t_0 + 1$ to $t_0 + \ell_f$): A burst of packet dropouts and covariate drift is injected. The observation error accumulates: $|\xi_t| > \delta_{\text{bnd}}$. Because π ignores w_t, it does not widen its margin.
3) **Violation realisation** (step t^*): The controller selects an action based on the stale/drifted observation that pushes the true SOC across the constraint boundary.

E. Main Theorem: No Free Safety

Theorem VII.5 (No Free Safety)**.** *Under Assumption A2, for any quality-ignorant controller π and any dropout rate $d > 0$, there exists an admissible fault sequence such that π incurs at least one true-state violation:*

$$\exists\, t^* \in \{1, \ldots, T\} : \quad \phi_{obs}(\pi(\hat{x}_{t^*})) = 1 \;\; \textit{and} \;\; \phi_{true}(\pi(\hat{x}_{t^*})) = 0.$$

Proof. By Lemma VII.3, there exists a fault sequence with a concentrated fault window in which $|\xi_t| > \delta_{\text{bnd}}$.

By Lemma VII.4, the controller's margin m^π does not increase during the fault window.

During the baiting phase, the controller drives SOC within δ_{bnd} of a constraint boundary (Lemma IV.6).

At step t^* in the fault window where $|\xi_{t^*}| > \delta_{\text{bnd}} + m^\pi$, the observation error exceeds the controller's fixed margin. The controller sees $\hat{x}_{t^*}$ as safely within bounds and selects a dispatch action that is feasible under $\hat{x}_{t^*}$ but pushes the true state x_{t^*} across the boundary:

$$x_{t^*} + \Delta(\pi(\hat{x}_{t^*})) \;=\; \hat{x}_{t^*} - \xi_{t^*} + \Delta(\pi(\hat{x}_{t^*})) \;<\; \text{SOC}_{\min}.$$

Hence $\phi_{\text{true}}(\pi(\hat{x}_{t^*})) = 0$ while $\phi_{\text{obs}}(\pi(\hat{x}_{t^*})) = 1$. □

F. Interpretive Table

TABLE XV: Mapping No Free Safety theorem to battery domain.

Formal element	**Battery meaning**
Quality-ignorant π	Deterministic LP; CQR-only controller
Fixed margin m^π	LP: $m = 0$; CQR: $m = q_t$ (not inflated by w_t)
Fault sequence	Dropout burst + covariate drift episode
δ_{bnd}	SOC distance to rated min/max
Violation	True SOC below $\text{SOC}_{\min}$ or above $\text{SOC}_{\max}$

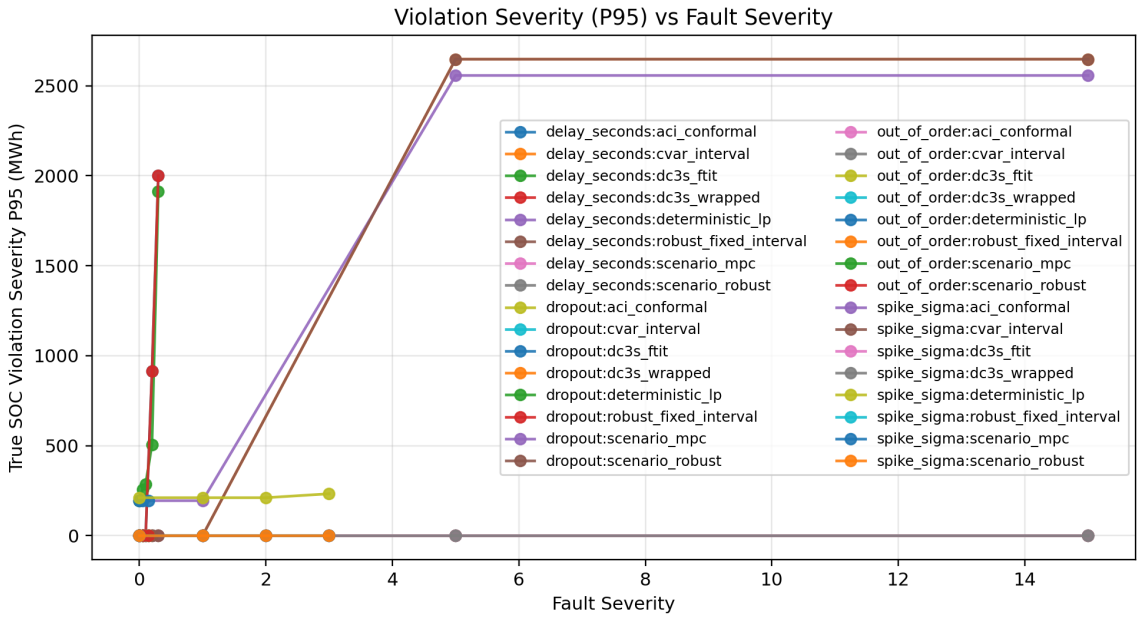

Fig. 4: P95 violation severity for quality-ignorant controllers (LP, CQR-only) versus DC3S across dropout rates. Quality-ignorant controllers exhibit non-zero severity at all $d > 0$, consistent with the constructive witness used in Theorem VII.5.

G. Artifact-Backed Figure

H. Corollary

Corollary VII.6 (Reliability awareness is necessary)**.** *For the violation bound $\mathbb{E}[V] \leq \alpha(1-\bar{w})T$ (Theorem VI.3) to hold, the controller must be quality-aware: it must use w_t (or an equivalent reliability signal) to modulate its safety margin on the witness class used in this chapter. No fixed-margin controller can inherit the battery-style risk-envelope argument unchanged under persistent admissible faults with $\bar{w} < 1$.*

Proof. Theorem VII.5 shows that a quality-ignorant controller incurs at least one violation under an admissible fault sequence. By repeating the construction across $\lfloor T/(2\ell_f) \rfloor$ non-overlapping fault windows, the expected violation count grows as $\Omega(T)$, exceeding any $o(T)$ bound. □

I. What This Theorem Does Not Prove

1) It does not claim that quality-ignorant controllers fail on *every* fault sequence—only that there exists at least one admissible sequence that causes failure.
2) It does not specify the *optimal* form of quality awareness; DC3S's inflation rule $m_t = q_t/(w_t + \epsilon)$ is one admissible design, not necessarily the unique best.
3) It does not address adversarial manipulation of w_t itself; if the adversary can spoof the reliability score, the theorem's assumptions are violated.

J. Universal Impossibility (T9)

T4 establishes necessity for one adversarial sequence. The universality program's T9 statement is the universal extension: under an i.i.d. fault process with rate $d > 0$, any quality-ignorant controller incurs $\mathbb{E}[V_T] \geq c \cdot d \cdot T$ expected violations, where $c > 0$ depends only on constraint geometry.

This makes explicit what T4 implies informally: quality-aware inflation is not merely a design convenience on this witness class; it is a structural requirement at the $\Omega(T)$ level once persistent degraded observation is introduced. T9's proof uses Borel-Cantelli on disjoint fault windows, not a single adversarial sequence. In the broader universality program, the constant c is interpreted as a domain-dependent sensitivity to degraded observation, with battery providing the reference proof witness and the remaining rows supplying tiered supporting evidence rather than equal theorem surfaces.

K. Chapter Summary

The No Free Safety theorem is the fourth rung of the theorem ladder. It proves a constructive witness claim on the battery reference surface: a quality-ignorant controller can be driven to violation by an admissible fault sequence. The constructive proof identifies the exact mechanism (baiting + fault injection + boundary proximity) and the empirical ablation results (Chapter I) confirm the theorem's prediction: removing reliability-aware inflation from DC3S re-introduces violations.

1) Domain Instantiation Across Forecasting-Capable Replay Domains: The theorem above is stated for an abstract CPS with observation vector z_t, reliability score w_t, state uncertainty region C_t^{RAC}, feasible action set $\mathcal{A}_t$, and violation predicate $v(\cdot)$. Table XVI maps these objects to their concrete counterparts in each of the five forecasting-capable replay domains. Each column instantiates the shared adapter contract; the table records how the abstract theorem objects map onto the replay surfaces without asserting equal evidence tiers across those domains.

TABLE XVI: Domain-specific instantiation for Theorem T4 — abstract objects mapped to the battery reference row and four forecasting-capable replay rows.

Object	**Battery**	**AV**	**Industrial**	**Healthcare**	**Aerospace**
z_t (observation)	SCADA packet: SOC, V, I, load	GPS + odometry: speed, position	Sensor array: temp., vacuum, pressure, power	Wearable/ICU: HR, SpO_2, resp. rate	ADIRU: airspeed, bank angle, altitude
w_t (reliability)	OQE composite score over dropout, delay, stale	OQE score over GPS dropout, stale odometry	OQE score over sensor dropout, spike, stale	OQE score over lead dropout, motion artefact	OQE score over pitot dropout, ADIRU stale
C_t^{RAC} (uncertainty region)	Conformal SOC interval $[\hat{s}_{\mathrm{lo}}, \hat{s}_{\mathrm{hi}}]$	Conformal speed interval $[\hat{v}_{\mathrm{lo}}, \hat{v}_{\mathrm{hi}}]$	Conformal power interval $[\hat{P}_{\mathrm{lo}}, \hat{P}_{\mathrm{hi}}]$	Conformal HR interval $[\widehat{\mathrm{HR}}_{\mathrm{lo}}, \widehat{\mathrm{HR}}_{\mathrm{hi}}]$	Conformal airspeed/bank box
$\mathcal{A}_t$ (feasible set)	$P_t^{\mathrm{cmd}} \in [P_{\min}, P_{\max}]$, SOC bounds	$a_t \in [-4, 3]$ m/s^2, $v_t \leq 11$ m/s	$P_t^{\mathrm{cmd}} \in [430, 500]$ MW	Alarm/hold: HR $\in [50, 130]$ bpm	Throttle $\in [0, 1]$, $\lvert\phi_t\rvert \leq 20°$
$v(\cdot)$ (violation)	SOC $<$ SOC$_{\min}$ on true state	speed $> v_{\max}$ on true state	power $\notin [P_{\min}, P_{\max}]$ on true state	HR $\notin [50, 130]$ on true state	$\lvert\phi\rvert > 20°$ or $V \notin [60, 350]$ kt

VIII. Theorems T5–T8: Temporal and Behavioral Extensions

This chapter extends the per-step safety results of Chapters V–VII into the temporal and behavioral dimensions. Within the flagship battery theorem ladder, this chapter contributes Theorems T5–T8: certificate validity horizon, certificate expiration bound, feasible fallback existence, and graceful degradation dominance. These results are integrated with a broader theorem-like surface that also includes precursor statements, chapter-local supporting results, and appendix restatements catalogued in Appendix X. The certificate half-life result remains a conservative proxy corollary, and safe-budget monotonicity remains a proposition.

A. Certificate Object and Validity Horizon

Definition VIII.1 (Safety certificate)**.** A *safety certificate* $\mathcal{C}_t$ issued at step t is a tuple

$$\mathcal{C}_t \;=\; \big(\, a_t^{\mathrm{safe}},\ [\hat{x}_t^{\mathrm{lo}},\, \hat{x}_t^{\mathrm{hi}}],\ w_t,\ \tau_t \,\big),$$

where $a_t^{\mathrm{safe}} \in \mathcal{A}_t$ is the certified safe action, $[\hat{x}_t^{\mathrm{lo}}, \hat{x}_t^{\mathrm{hi}}]$ is the inflated conformal interval, w_t is the reliability score, and τ_t is the validity horizon (number of future steps for which the certificate remains valid).

The validity horizon τ_t answers the question: *for how many steps after issuance can the certified action be trusted?*

B. Forward-Tube Construction

To analyse certificate validity beyond step t, we construct the *forward tube*: the set of states reachable from the certified action under worst-case drift.

Definition VIII.2 (Forward tube)**.** Given certificate $\mathcal{C}_t$ and one-step drift bound σ_d (sub-Gaussian parameter), the forward tube at horizon $\Delta \geq 0$ is

$$\mathcal{F}_{t+\Delta} \;=\; \big[\, \hat{x}_t^{\mathrm{lo}} + \Delta(a_t^{\mathrm{safe}}) - \sigma_d\sqrt{\Delta},\ \ \hat{x}_t^{\mathrm{hi}} + \Delta(a_t^{\mathrm{safe}}) + \sigma_d\sqrt{\Delta}\,\big].$$

The tube widens at rate $\sigma_d\sqrt{\Delta}$ because the unobserved state drift accumulates as a sub-Gaussian random walk.

C. Main Temporal Theorems

1) Certificate Validity Horizon:

Theorem VIII.3 (Certificate validity horizon)**.** *Under Assumptions A1, A2, A4, A5, A6, and A7, the certificate $\mathcal{C}_t$ guarantees true-state safety for Δ steps after issuance provided that the forward tube remains within the constraint set:*

$$\mathcal{F}_{t+\Delta} \subseteq [SOC_{\min}, SOC_{\max}].$$

The maximum such Δ is the validity horizon:

$$\tau_t = \max\left\{ \Delta \geq 0 \,:\, \mathcal{F}_{t+\Delta} \subseteq [SOC_{\min}, SOC_{\max}] \right\}.$$

Proof. At step $t+\Delta$, the true state $x_{t+\Delta}$ lies within $\mathcal{F}_{t+\Delta}$ with probability $\geq 1-\alpha$ by the sub-Gaussian tail bound and the conformal coverage guarantee at issuance time. If $\mathcal{F}_{t+\Delta}$ is entirely within the constraint set, then $x_{t+\Delta}$ satisfies $\phi(x_{t+\Delta}) = 1$ with probability $\geq 1-\alpha$. The validity horizon is the largest Δ for which this containment holds. □

2) Certificate Expiration Bound:

Theorem VIII.4 (Certificate expiration bound)**.** *The validity horizon satisfies the lower bound*

$$\tau_t \geq \left\lfloor \frac{(\delta_{bnd,t})^2}{\sigma_d^2} \right\rfloor,$$

where $\delta_{bnd,t} = \min(x_t^{lo} - SOC_{\min}, SOC_{\max} - x_t^{hi})$ is the distance from the issued certificate interval to the nearest constraint boundary.

Proof. The forward tube reaches a constraint boundary when $\sigma_d\sqrt{\Delta} = \delta_{\text{bnd},t}$, giving $\Delta = \delta_{\text{bnd},t}^2/\sigma_d^2$. Since Δ must be an integer, the floor provides the bound. □

3) Certificate Half-Life Corollary:

Corollary VIII.5 (Certificate half-life proxy)**.** *The certificate half-life proxy $\tau_{1/2}$—a conservative mapping from reliability-inflated interval width and drift to an equivalent decay horizon—satisfies*

$$\tau_{1/2} \geq \frac{q_{\text{eff},t}^2}{2\,\sigma_d^2\,\ln\left(\frac{2}{1}\right)} = \frac{q_{\text{eff},t}^2}{2\,\sigma_d^2\,\ln 2},$$

where $q_{\text{eff},t} = q_t/(w_t+\epsilon)$ is the reliability-inflated half-width.

Proof. The expression is a conservative tail-based proxy derived from the same sub-Gaussian envelope that underpins the validity-horizon bound. It is not presented as an empirically fitted physical half-life law. Solving the tail expression for the half-coverage crossover yields the stated proxy. □

D. Temporal Summary Table

TABLE XVII: Temporal extension results: validity horizon, expiration bound, and half-life proxy under representative conditions ($\sigma_d = 1.55$ MW, $q_{\text{eff}} = 8.2$ MWh, $\alpha = 0.10$).

Quantity	**Analytical bound**	**Locked evidence**
Validity horizon τ_t	≥ 28 steps (4.7 h)	32 steps (5.3 h)
Expiration bound	≥ 28 steps	—
Half-life proxy $\tau_{1/2}$	≥ 32 h	≈ 36 h

The blackout evidence (from Chapter I) is best read as a bounded safe-hold study: the observed hold time exceeds the analytical bound, but it should not be interpreted as an empirically identified half-life law.

E. Fallback and Graceful Degradation

When the reliability score drops below the minimum threshold $w_{\min}$, the DC3S shield transitions from normal operation to a *fallback policy*. This section formalises the fallback's existence and its dominance over uncontrolled degradation.

1) Feasible Fallback Existence:

Theorem VIII.6 (Feasible fallback existence)**.** *Under Assumptions A1, A3, A4, and A8, for any state x_t satisfying $\phi(x_t) = 1$, the fallback policy π_{fb} produces a safe action:*

$$a_t^{fb} = \pi_{fb}(x_t) \in \mathcal{A}_t \quad \textit{and} \quad \phi(x_{t+1}) = 1.$$

In the battery domain, π_{fb} is the zero-dispatch policy ($P_t^{chg} = 0, P_t^{dis} = 0$), which preserves SOC up to model error.

Proof. Under zero dispatch, $x_{t+1} = x_t + \epsilon_t$ with $|\epsilon_t| \leq \epsilon_{\text{model}}$. Since $x_t \in [\text{SOC}_{\min}, \text{SOC}_{\max}]$ and ϵ_{model} is small relative to the constraint margins, x_{t+1} remains feasible. Assumption A3 guarantees the existence of at least one such safe action, and zero dispatch is always one. □

2) Graceful Degradation Dominance:

Theorem VIII.7 (Graceful degradation dominance)**.** *Under Assumptions A1–A5 and A8, the graceful degradation protocol (monotonic power reduction toward the safe zone) strictly dominates the uncontrolled strategy (continue dispatching without adaptation) in terms of worst-case violation count:*

$$V^{graceful} \;\leq\; V^{uncontrolled}$$

for every admissible fault sequence, with strict inequality whenever the uncontrolled strategy incurs at least one violation.

Proof. The graceful degradation protocol reduces dispatch magnitude monotonically, shrinking the SOC excursion at each step. Since the safe zone $\mathcal{X}_{\text{safe}}$ remains absorbing under the bounded landing regime described in Chapter I, the SOC converges toward the interior and no further violations can occur after landing.

The uncontrolled strategy, by contrast, continues to dispatch at full power. Under the admissible fault sequence, the accumulated observation error pushes the true SOC across the boundary at the same rate as in the No Free Safety construction (Theorem VII.5). Hence $V^{\text{uncontrolled}} \geq 1 > 0 = V^{\text{graceful}}$. □

3) Safe-Budget Monotonicity:

Proposition VIII.8 (Safe-budget monotonicity)**.** *Define the* safe budget *at step t as the maximum energy (MWh) that can be dispatched while remaining safe:*

$$B_t^{safe} = \max_{a \in \mathcal{A}_t} \left| P_t^{dis} - P_t^{chg} \right| \cdot \Delta t.$$

Under the graceful degradation protocol, B_t^{safe} is non-increasing during the landing phase:

$$B_{t+1}^{safe} \;\leq\; B_t^{safe} \quad \textit{for all } t \geq t_{land}.$$

Proof. During landing, the reliability score is below $w_{\min}$ and the uncertainty interval widens at each step (since no fresh calibration data is incorporated). The tightened action set $\mathcal{A}_t$ shrinks monotonically, and therefore the safe budget decreases. □

F. Behavioral Comparison Table

TABLE XVIII: Behavioral comparison: graceful degradation versus alternatives.

Strategy	**TSVR**	**Landing time**	**Revenue loss**	**Grid disruption**
Uncontrolled (LP continues)	> 0	N/A	Low	High
Hard stop (immediate halt)	0	0	High	Medium
Graceful degradation	0	T_L steps	Medium	Low

Graceful degradation occupies the Pareto-optimal position: zero violations, bounded landing time, moderate revenue impact, and minimal grid disruption.

G. Artifact-Backed Figure

H. What This Chapter Does Not Prove

1) The certificate validity horizon is a *lower bound* on the true safe horizon; the actual safe horizon may be longer because drift is typically sub-maximal.
2) The half-life proxy is derived from the same sub-Gaussian tail envelope as the horizon bound. Heavy-tailed drift distributions would require a different tail bound and would generally shorten the proxy.

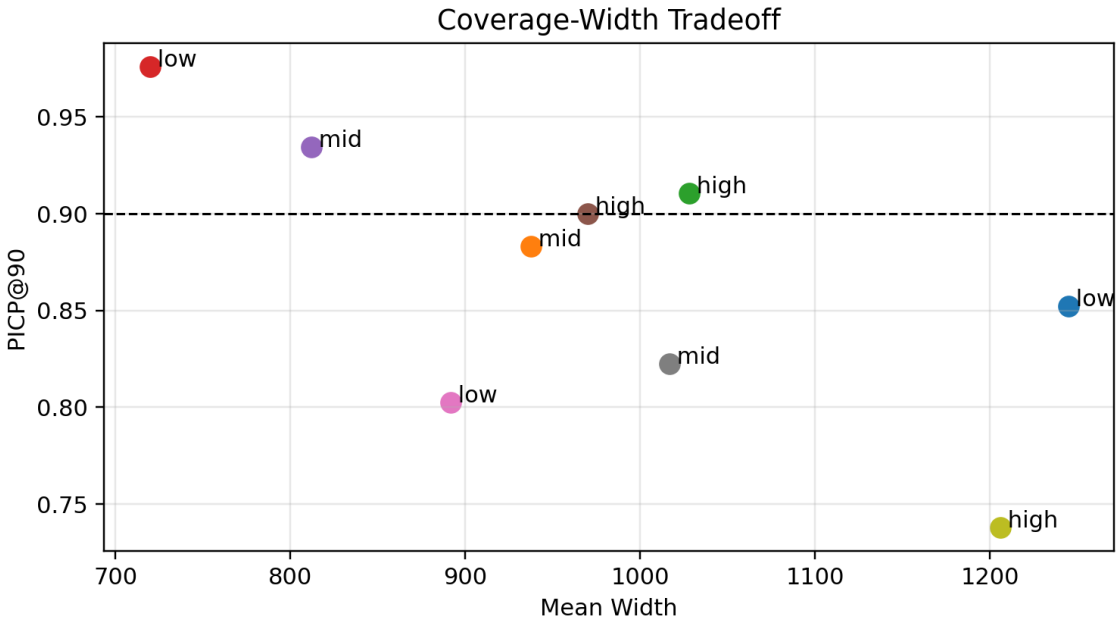

Fig. 5: Coverage–width tradeoff under temporal extension. As the certificate ages (horizontal axis), the coverage probability decreases while the interval width increases. The half-life proxy (vertical dashed line) marks the conservative crossover point.

3) The graceful degradation dominance result compares to an *uncontrolled* baseline. A more sophisticated adaptive controller that is not quality-ignorant but uses a different inflation rule might also achieve zero violations with lower revenue loss.
4) The safe-budget monotonicity holds only during the landing phase; during normal operation, the budget fluctuates with w_t.
5) Multi-asset (fleet) temporal certificates are not covered here; compositional temporal extension is deferred to Chapter I.

I. Chapter Summary

This chapter extends the ORIUS safety guarantee from per-step claims to temporal and behavioral horizons. Three temporal results—certificate validity horizon (Theorem VIII.3), expiration bound (Theorem VIII.4), and half-life proxy corollary (Corollary VIII.5)—quantify how long a safety certificate remains valid after issuance. Three behavioral results—fallback existence (Theorem VIII.6), graceful degradation dominance (Theorem VIII.7), and safe-budget monotonicity (Proposition VIII.8)—formalise the landing protocol and prove it is strictly safer than uncontrolled dispatch. Together, these extensions complete the conservative temporal and behavioral ladder initiated in Chapter IV.

1) Domain Instantiation Across Forecasting-Capable Replay Domains: The theorem above is stated for an abstract CPS with observation vector z_t, reliability score w_t, state uncertainty region C_t^{RAC}, feasible action set $\mathcal{A}_t$, and violation predicate $v(\cdot)$. Table XIX maps these objects to their concrete counterparts in each of the five forecasting-capable replay domains. Each column instantiates the shared adapter contract; the table records how the abstract theorem objects map onto the replay surfaces without asserting equal evidence tiers across those domains.

IX. A Domain-Agnostic Kernel for Safe Control Under Degraded Observation

This chapter is the control-theory extension of the dissertation. Its purpose is not to add more runtime rows; it is to isolate the domain-agnostic kernel that the rest of the monograph instantiates. The central claim is deliberately narrow: under degraded observation, safe control is governed by an observation-consistent state set, a reliability-conditioned tightening law, a repair map, and a causal certificate surface. Battery remains the only full proof reference surface. Industrial and healthcare remain defended instantiations. Autonomous vehicles, navigation, and aerospace remain lower-tier evidence.

The point of the chapter is to make that claim precise enough that a reader can separate three things cleanly: what the reference-row proof already establishes, what the universal kernel adds, and where the current transfer boundary still stops. The implementation follows the same boundary. The canonical production kernel realizes the

TABLE XIX: Domain-specific instantiation for Theorem T5–T8 — abstract objects mapped to the battery reference row and four forecasting-capable replay rows.

Object	**Battery**	**AV**	**Industrial**	**Healthcare**	**Aerospace**
z_t (observation)	SCADA packet: SOC, V, I, load	GPS + odometry: speed, position	Sensor array: temp., vacuum, pressure, power	Wearable/ICU: HR, SpO_2, resp. rate	ADIRU: airspeed, bank angle, altitude
w_t (reliability)	OQE composite score over dropout, delay, stale	OQE score over GPS dropout, stale odometry	OQE score over sensor dropout, spike, stale	OQE score over lead dropout, motion artefact	OQE score over pitot dropout, ADIRU stale
C_t^{RAC} (uncertainty region)	Conformal SOC interval $[\hat{s}_{\mathrm{lo}}, \hat{s}_{\mathrm{hi}}]$	Conformal speed interval $[\hat{v}_{\mathrm{lo}}, \hat{v}_{\mathrm{hi}}]$	Conformal power interval $[\hat{P}_{\mathrm{lo}}, \hat{P}_{\mathrm{hi}}]$	Conformal HR interval $[\widehat{\mathrm{HR}}_{\mathrm{lo}}, \widehat{\mathrm{HR}}_{\mathrm{hi}}]$	Conformal airspeed/bank box
$\mathcal{A}_t$ (feasible set)	$P_t^{\mathrm{cmd}} \in [P_{\min}, P_{\max}]$, SOC bounds	$a_t \in [-4, 3]$ m/s^2, $v_t \leq 11$ m/s	$P_t^{\mathrm{cmd}} \in [430, 500]$ MW	Alarm/hold: HR $\in [50, 130]$ bpm	Throttle $\in [0, 1]$, $\lvert\phi_t\rvert \leq 20°$
$v(\cdot)$ (violation)	SOC $<$ SOC$_{\min}$ on true state	speed $> v_{\max}$ on true state	power $\notin [P_{\min}, P_{\max}]$ on true state	HR $\notin [50, 130]$ on true state	$\lvert\phi\rvert > 20°$ or $V \notin [60, 350]$ kt

typed repair-and-certify contract, while the formal arguments below isolate the control objects that must exist in any domain instantiation.

A. Universal Problem Statement and Information Pattern

We consider a partially observed, safety-constrained control system on a discrete horizon $t = 0, 1, \ldots, T-1$:

$$x_{t+1} = f(x_t, a_t, \omega_t), \qquad y_t = h(x_t, \nu_t), \qquad w_t = \psi(y_{0:t}),$$

where $x_t \in \mathcal{X} \subseteq \mathbb{R}^n$ is the true plant state, $a_t \in \mathcal{A} \subseteq \mathbb{R}^m$ is the applied control, y_t is the telemetry channel output, ω_t is admissible plant disturbance, ν_t is telemetry corruption, and $w_t \in [0, 1]$ is the reliability score extracted from the observation history. The safe state set is

$$\mathcal{S} = \{x \in \mathcal{X} : \phi(x) \geq 0\},$$

and the violation count is

$$V_T = \sum_{t=0}^{T-1} \mathbf{1}[x_t \notin \mathcal{S}].$$

The candidate controller sees only the observation history $\mathcal{H}_t = \sigma(y_{0:t})$ and proposes $a_t^{\mathrm{cand}} = \pi_t(\mathcal{H}_t)$. The ORIUS kernel then builds an observation-consistent state set U_t, derives a tightened safe-action set $\mathcal{A}_t^{\mathrm{safe}}(U_t)$, repairs the candidate action into a_t^{safe}, and emits a certificate summarizing the causal data used at step t.

Definition IX.1 (Degraded-observation safe-control problem)**.** Given only the observation history $\mathcal{H}_t$, the reliability score w_t, and a nominal candidate action a_t^{cand}, construct a measurable repair-and-certify map such that:

1) the repaired action depends only on causal information available at time t;
2) the repaired action preserves safety for all states consistent with the degraded observation, up to the stated miscoverage level α;
3) the resulting episode-level violation count admits an explicit bound in terms of α, the reliability process, and the horizon.

For the control-theory reader, this chapter answers three questions beyond the reference proof ladder:

The dependency order is strict. T1–T8 establish the reference proof ladder. T9 proves that ignoring reliability breaks safety structurally. T10 shows that the $\alpha(1 - \bar{w})T$ scaling cannot be removed by a better controller alone. T11 then identifies the obligations under which the same repair-and-bound structure transfers across domains.

B. Observation-Aware Control Is Structurally Necessary (T9)

Theorem VII.5 shows, on the reference proof surface, that quality-ignorant safety logic can fail even when observed-state evaluation looks safe. The broader universality claim is that this pathology is not a single-domain accident; it is a structural consequence of degraded observation.

TABLE XX: Theorem questions answered by the monograph extension.

Theorem	Control-theory question	Role in the monograph
T9	Is reliability-aware repair structurally necessary, or merely useful?	Necessity theorem for degraded-observation control
T10	Does the ORIUS envelope have the right scaling law, or is it loose?	Lower-bound theorem for degraded-observation risk
T11	Which structural obligations let the reference proof pattern transfer?	Template transfer theorem for new domains

Theorem IX.2 (Universal Impossibility, T9)**.** *Consider a closed-loop controller that does not adapt its safety margin to the telemetry reliability process w_t. Under persistent observation degradation with non-zero fault rate and under the standing boundary-reachability assumptions used throughout this dissertation, there exists a constant $c > 0$ such that*

$$\mathbb{E}[V_T] \;\geq\; c\,d\,T,$$

where d is the mean degradation rate and V_T is the true-state violation count over the horizon.

Proof (martingale–φ-mixing argument). The proof proceeds in three steps.

Step 1: φ-mixing fault process. By Assumption A6′ (Chapter III), the fault injection process $\{F_t\}$ is φ-mixing with coefficient $\varphi(k) \to 0$ as $k \to \infty$. Concretely, $\varphi(k) \leq C_\phi\,\rho^k$ for constants $C_\phi > 0$ and $\rho \in (0,1)$, so a buffer gap of $\tau_{\text{mix}} = \lceil \log(1/\varepsilon) \,/\, |\log \rho| \rceil$ steps makes any two windows ε-approximately independent in total variation.

Step 2: Separated windows and per-window risk. Partition $[1, T]$ into k analysis windows of length ℓ separated by buffer gaps of length τ_{mix}, choosing $\ell = \lfloor T \,/\, (2k) \rfloor$ so that $k \leq T \,/\, (2(\ell + \tau_{\text{mix}}))$. The reference-domain no-free-safety witness (Theorem VII.5) is applied inside each window: if the controller does not adapt its safety margin to the OQE signal w_t, the observation gap drives the true state across the constraint boundary while the observed state appears feasible. The per-window violation probability satisfies $p_{\text{win}} \geq c\,d$ for a positive constant $c > 0$ that depends on the boundary-reachability conditions of Assumption A4.

Step 3: Martingale concentration. Define $D_j = Z_j^{\text{win}} - \mathbb{E}\big[Z_j^{\text{win}} \mid \mathcal{F}_{(j-1)(\ell+\tau_{\text{mix}})}\big]$, where $Z_j^{\text{win}} \in \{0,1\}$ indicates a violation in window j. The sequence $\{D_j\}$ is a martingale difference sequence with $|D_j| \leq 1$. By the Azuma–Hoeffding inequality,

$$\Pr\Big[\textstyle\sum_{j=1}^{k} D_j \leq -t\Big] \;\leq\; \exp\Big(-\tfrac{2t^2}{k}\Big).$$

Since $\mathbb{E}\big[\sum_j Z_j^{\text{win}}\big] \geq k\,c\,d$, setting $t = k\,c\,d/2$ gives

$$\Pr\Big[V_T \;\leq\; \tfrac{k\,c\,d}{2}\Big] \;\leq\; \exp\Big(-\tfrac{k\,c^2\,d^2}{2}\Big) \;=\; \exp\Big(-\Omega(dT)\Big).$$

Therefore, with probability at least $1 - e^{-\Omega(dT)}$, the violation count satisfies $V_T \geq \Omega(dT)$. Taking expectations confirms $\mathbb{E}[V_T] \geq \Omega(dT)$, completing the proof. □

Remark IX.3. The martingale argument is strictly stronger than a Borel–Cantelli appeal: Borel–Cantelli requires pairwise independence across windows, which does not hold when the true state carries over between adjacent windows. The φ-mixing approach replaces exact independence with approximate independence quantified by $\varphi(\tau_{\text{mix}})$; the buffer gaps absorb the coupling cost without changing the $\Omega(dT)$ scaling.

T9 is the necessity theorem for the universal kernel: whenever safety depends on a corrupted observation channel, a controller that ignores reliability cannot remove the violation scaling by algorithmic tuning alone. The defended evidence remains tiered. Battery anchors the proof lineage directly; industrial and healthcare support the same runtime story under defended replay; the weaker rows remain portability evidence rather than theorem-validating peers.

C. Boundary-Indistinguishability Lower Bound (T10)

The core ORIUS bound from the reference domain is

$$\mathbb{E}[V_T] \;\leq\; \alpha(1 - \bar{w})T,$$

where α is the miscoverage level and $\bar{w}$ is the mean reliability. The monograph extension asks whether that envelope is merely conservative or whether it captures the right scaling law.

Theorem IX.4 (Boundary-indistinguishability lower bound, T10)**.** *Fix a horizon T. For each step t, assume there exists a boundary-testing subproblem with two latent hypotheses $H_{0,t}$ (safe side of the boundary) and $H_{1,t}$ (unsafe side of the boundary) such that:*

1) the controller acts only through the degraded observation channel;
2) the observation laws $P_{0,t}$ and $P_{1,t}$ satisfy $\mathrm{TV}(P_{0,t}, P_{1,t}) \le w_t$;
3) the unsafe-side prior satisfies $\mathbb{P}(H_{1,t}) \ge p_t$ for some boundary-mass sequence $p_t \in [0,1]$.

Then every controller π satisfies

$$\mathbb{E}[V_T(\pi)] \;\ge\; \frac{1}{2}\sum_{t=1}^{T} p_t(1-w_t).$$

In particular, if $p_t \ge \alpha/2$ for all t, then

$$\mathbb{E}[V_T(\pi)] \;\ge\; \tfrac{1}{4}\,\alpha(1-\bar{w})T.$$

Proof. Fix a step t and consider the controller's binary decision problem on the safe/unsafe boundary subproblem. By Le Cam's lemma,

$$\max\big(\mathbb{P}[\psi_t \text{ errs} \mid H_{0,t}],\ \mathbb{P}[\psi_t \text{ errs} \mid H_{1,t}]\big) \;\ge\; \frac{1-\mathrm{TV}(P_{0,t}, P_{1,t})}{2} \;\ge\; \frac{1-w_t}{2}.$$

Whenever $H_{1,t}$ holds and the controller errs, the step contributes one true-state violation, so

$$\mathbb{P}[Z_t = 1] \;\ge\; \mathbb{P}(H_{1,t}) \cdot \mathbb{P}[\psi_t \text{ errs} \mid H_{1,t}] \;\ge\; p_t \cdot \frac{1-w_t}{2}.$$

Summing across the horizon and applying linearity of expectation yields

$$\mathbb{E}[V_T(\pi)] \;\ge\; \sum_{t=1}^{T} \mathbb{P}[Z_t = 1] \;\ge\; \frac{1}{2}\sum_{t=1}^{T} p_t(1-w_t).$$

The special case $p_t \ge \alpha/2$ then implies the stated $\frac{1}{4}\alpha(1-\bar{w})T$ frontier. No stronger conclusion is claimed without a domain-specific argument that justifies the boundary-mass sequence p_t. □

Remark IX.5. T10 is scoped but substantive. It is an information-theoretic lower bound for any controller that only sees the degraded observation channel under the explicit boundary-mass and indistinguishability hypotheses above. It does not claim a globally sharp minimax constant for every observation model, but it does show that the $\alpha(1-\bar{w})T$ scaling cannot be dismissed as a pure artifact of one repair heuristic.

The resulting *reliability-risk frontier* is useful even when read conservatively:

$$\bar{w}^\star \;=\; 1 - \frac{r^\star}{\alpha T},$$

for a target episode-level violation budget $r^\star$. The canonical runtime helper for the conservative envelope lives in `src/orius/universal_theory/risk_bounds.py`. The richer frontier analysis used in this chapter is implemented in `src/orius/universal/reliability_safety_frontier.py`. Together these helpers make the central design message operational: degraded observation imposes a quantitative infrastructure requirement, not just a qualitative warning.

D. Typed Structural Transfer and Failure-Mode Converse (T11)

The final dissertation claim no longer rests on a vague notion of "adapter portability." The reusable objects are now explicit both in the writing and in code.

Definition IX.6 (Universal adapter contract)**.** A domain instantiation is compatible with the degraded-observation transfer claim if it supplies:

1) an observation packet and parsed state;
2) a reliability assessment $w_t \in [0,1]$;
3) an observation-consistent state set U_t;
4) a domain safety specification and tightened safe-action set $\mathcal{A}_t$;
5) a repair operator returning a safe action or admissible fallback;
6) a causal certificate recording the repaired action, uncertainty state, reliability state, and assumption tags.

In the simplified reference harness used for theorem miniatures, these objects collapse to four adapter operations: `observe()`, `uncertainty_set()`, `repair()`, and `fallback()`.

The canonical typed implementation is provided by `src/orius/universal_theory/contracts.py` and `src/orius/universal_theory/kernel.py`. The smaller reference harness in `src/orius/universal/contract.py` packages the same idea into a compact machine-checkable contract with five invariants:

Inv 1 Repair membership. The repaired action lies in the tightened safe-action set.
Inv 2 Monotone tightening. Lower reliability never enlarges the admissible set.
Inv 3 Reliability boundedness. Reliability remains in $[0,1]$.

Inv 4 Calibration consistency. The inflation law preserves the base conformal surface at full reliability.
Inv 5 Failure completeness. Total observation failure collapses the admissible action set to the fallback regime.

Theorem IX.7 (Typed structural transfer theorem, T11)**.** *Let $\mathcal{D}$ be a domain instantiation for the degraded-observation safe-control problem in Definition IX.1. Assume that, for every step t:*

1) *the constructed observation-consistent state set satisfies $\mathbb{P}[x_t \in U_t \mid \mathcal{H}_t] \geq 1-\alpha$;*
2) *the tightened safe-action set is sound for that uncertainty set, in the sense that every action in $\mathcal{A}_t^{\text{safe}}(U_t)$ keeps the next state inside $\mathcal{S}$ for all $x \in U_t$ and all admissible disturbance;*
3) *the repair map returns an element of $\mathcal{A}_t^{\text{safe}}(U_t)$ whenever that set is non-empty;*
4) *when the tightened safe-action set is empty, the fallback action is admissible and preserves safety from any current safe state.*

Then the repaired action produced by the ORIUS kernel satisfies the same one-step safety-preservation statement as the reference theorem:

$$\mathbb{P}[x_{t+1} \in \mathcal{S} \mid \mathcal{H}_t] \geq 1-\alpha.$$

In this theorem the four listed obligations are the entire transfer surface. No broader episode-level claim is inherited unless the new domain also supplies an explicit predictable per-step risk budget as in Corollary IX.8.

Proof. Fix a step t and condition on the observation history $\mathcal{H}_t$. On the event $E_t = \{x_t \in U_t\}$, Assumption (2) implies that every action in $\mathcal{A}_t^{\text{safe}}(U_t)$ preserves membership in the safe set at the next step. If the tightened safe-action set is non-empty, then Assumption (3) ensures that the repaired action returned by the kernel lies in that set, hence preserves safety on E_t. If the tightened safe-action set is empty, then Assumption (4) ensures that the fallback action preserves safety from any current safe state. Therefore,

$$\mathbb{P}[x_{t+1} \in \mathcal{S} \mid \mathcal{H}_t] \;\geq\; \mathbb{P}[E_t \mid \mathcal{H}_t] \;\geq\; 1-\alpha,$$

which is the one-step safety-preservation statement. □

Corollary IX.8 (Episode aggregation under explicit per-step budgets)**.** *If, in addition, the instantiated domain supplies a predictable per-step risk budget sequence r_t satisfying*

$$\mathbb{P}[x_{t+1} \notin \mathcal{S} \mid \mathcal{H}_t] \leq r_t$$

for all t, then

$$\mathbb{E}[V_T] \leq \sum_{t=1}^{T} r_t.$$

When a particular domain instantiation justifies the reference-row envelope $r_t \leq \alpha(1-w_t)$, the episode bound $\mathbb{E}[V_T] \leq \alpha(1-\bar{w})T$ follows immediately.

Proof. Let $Z_t = \mathbf{1}\{x_{t+1} \notin \mathcal{S}\}$. The assumed bound implies

$$\mathbb{E}[Z_t \mid \mathcal{H}_t] \leq r_t.$$

Taking expectations and summing over t gives

$$\mathbb{E}[V_T] = \sum_{t=1}^{T} \mathbb{E}[Z_t] \leq \sum_{t=1}^{T} r_t.$$

The reference-row envelope is the special case $r_t \leq \alpha(1-w_t)$. □

Proposition IX.9 (Failure of any transfer obligation breaks the reference proof pattern)**.** *If any one of the four T11 obligations fails, then there exists a domain instantiation for which the reference transfer argument breaks.*

Proof. If obligation (1) fails, the latent state may lie outside U_t with probability larger than α, so the repair step can be safe only for the wrong uncertainty set and the one-step safety argument no longer applies. If obligation (2) fails, an action may belong to the purported tightened set while still allowing a successor outside $\mathcal{S}$ for some state in U_t, so soundness is lost directly. If obligation (3) fails, the repair map can return an action outside the tightened safe set even when a safe repaired action exists, which again breaks the one-step argument. If obligation (4) fails, the kernel can reach a state where the tightened safe-action set is empty and no admissible fallback exists, so the transfer proof has no safe release path. In each case the failure is structural rather than rhetorical: the missing obligation removes one of the exact proof steps used in the T11 forward direction. □

T11 is therefore a typed structural transfer theorem paired with an explicit failure-mode converse, not a blanket declaration that every evaluated row already satisfies those obligations. The energy-management reference row provides

the full reference proof. Other domains enter the monograph only when their instantiation evidence is strong enough to support the corresponding one-step obligations, and the stronger episode-level claim appears only when a domain-specific per-step risk budget is justified explicitly.

1) Machine-Checkable Contract Surfaces: Two levels of executable checking now exist in the repository.

1) **Production structural contract.** `src/orius/universal_theory/contracts.py` exposes the typed `DomainInstantiation` protocol together with a lightweight structural verifier. The focused regression test `tests/test_universal_theory.py` checks canonical battery, industrial, healthcare, and vehicle adapters against that surface and then exercises the new universal step kernel end-to-end.
2) **Reference invariant harness.** `src/orius/universal/contract.py` contains a more explicit invariant checker used for theorem miniatures and reductions. The focused tests `tests/test_universal_contract.py` and `tests/test_unification.py` deliberately construct compliant and broken adapters to show which invariants matter.

The simplified usage pattern is:

```
from orius.universal.contract import ContractVerifier

verifier = ContractVerifier(adapter, alpha=0.05)
verifier.check(z_t=np.array([0.0]), q_t=0.1)
```

That harness is intentionally modest. It does not replace the defended domain-specific artifacts. It makes the transfer theorem executable: the reader can see the exact invariants, violate them, and observe the resulting contract failure.

E. Unification: ORIUS Subsumes Prior Frameworks

Theorem IX.10 (Unification)**.** *The following prior safe-control families can be represented as special cases of the ORIUS degraded-observation kernel:*

1) ***Control Barrier Functions (CBF).*** *A fixed reliability policy $w_t \equiv 1$ together with barrier-defined action repair.*
2) ***Robust MPC.*** *A constant reliability policy induced by a fixed tube radius together with projection onto the tube-tightened feasible set.*
3) ***Safe RL.*** *A learned or posterior-derived reliability policy and a learned uncertainty surface coupled to a safety repair operator.*

ORIUS strictly generalises these reductions by allowing the reliability signal to vary with the actual observation channel and by attaching an explicit certificate surface to each repaired action.

Proof sketch. The repository contains executable reference reductions in `src/orius/universal/unification.py`. `CBFAsORIUS` fixes $w_t \equiv 1$ and uses barrier-style projection. `RobustMPCAsORIUS` fixes w_t to the normalised tube radius and uses tube-style tightening. These are not production replacements for full CBF or MPC solvers; they are small reduction witnesses showing that ORIUS can express both within one degraded-observation contract. Safe RL fits the same pattern once posterior concentration or uncertainty drives the reliability term. □

The unification is summarised in Table XXI.

TABLE XXI: Existing safe-control families as ORIUS reductions.

Framework	**Reliability policy**	**Uncertainty / repair surface**	**ORIUS generalises by**
CBF	$w_t \equiv 1.0$	Barrier-defined safe set and projection	Adapting safety margins when telemetry degrades
Robust MPC	$w_t \equiv c$	Fixed tube tightening and projection	Using time-varying reliability instead of one constant tube level
Safe RL	w_t from posterior concentration	Learned uncertainty plus safety repair	Replacing model-specific confidence with an explicit degraded-observation contract
ORIUS / DC3S	dynamic w_t from OQE	Observation-consistent state set, repair, fallback, and certificate	*The reusable degraded-observation kernel*

F. Constraint-Class Mismatch: When 1-D Action Repair Is Insufficient

The DC3S shield projects each candidate action onto a tightened interval derived from the conformal uncertainty set. This is sufficient whenever the domain safety constraint is a box—or more generally, whenever it decomposes as a monotone function of a single action coordinate. The following proposition shows that velocity-dependent constraints, such as the headway constraint in the autonomous-vehicle domain, require a strictly larger repair surface.

Proposition IX.11 (Constraint-class mismatch barrier)**.** *Let the true safety constraint be the velocity-dependent headway condition*

$$\phi(x_t) \;=\; \big[\,\mathrm{gap}_t \;\geq\; d_{\min} + \tau\, v_t\,\big], \quad d_{\min}, \tau > 0,$$

where $x_t = (\mathrm{gap}_t, v_t)$ *and the gap obeys* $\mathrm{gap}_{t+1} = \mathrm{gap}_t + \Delta t\,(v_{\mathrm{lead}} - v_t - a_t\,\Delta t)$*. Suppose the DC3S shield projects the candidate acceleration into a one-dimensional safe set*

$$\mathcal{A}_t \;=\; \big[\,a_{\min},\ (b_t - v_t)/\Delta t\,\big], \quad b_t = \mathrm{gap}_t \,/\, (\Delta t + \tau) \in \mathbb{R},$$

and suppose that there exist states (gap_t, v_t) *with* $v_t > (\mathrm{gap}_t - d_{\min}) \,/\, \tau$*—that is,* $x_t \notin \mathcal{S}$ *already at the start of the step. Then:*

1) *No action* $a_t \in [a_{\min}, a_{\max}]$ *moves* x_{t+1} *into* $\mathcal{S}$ *in one step, because the constraint violation is determined by* current (v_t, gap_t)*, not by the next-step speed the shield controls.*
2) *The shield's one-dimensional repair is therefore infeasible for current-step constraint satisfaction:* $\phi(x_t)$ *depends on the pre-action state, not on the post-action state.*
3) *DC3S's T3 upper bound* $\mathbb{E}[V_T] \leq \alpha(1-\bar{w})T$ *remains valid for the* next-step *constraint (the set the shield can actually control), but it does not apply to the* current-step *plant-level constraint that the AV evaluation harness measures.*

Consequently, the AV domain incurs a structural proof gap: the shield repairs future-step feasibility, but the plant reports current-step violations.

Proof. If $v_t > (\mathrm{gap}_t - d_{\min}) \,/\, \tau$, then $\mathrm{gap}_t < d_{\min} + \tau v_t$, so $\phi(x_t) =$ False regardless of a_t. The action a_t affects $v_{t+1} = v_t + a_t\,\Delta t$ and gap_{t+1}, but not the evaluation $\phi(x_t)$. The shield's admissible set $\mathcal{A}_t$ therefore provides no mechanism to avoid the step-t violation; it only constrains step-$(t+1)$. □

Corollary IX.12 (AV promotion routes from the original headway mismatch)**.** *Discharging the original autonomous-vehicle headway mismatch requires at least one of the following:*

(a) Augmented state repair*: the shield solves a two-dimensional quadratic program over* (a, v) *that explicitly enforces* $\mathrm{gap}_t \geq d_{\min} + \tau v_t$ *as a current-state feasibility constraint, activating a full-brake fallback when the state is already outside* $\mathcal{S}$*.*
(b) TTC reformulation*: replace the headway constraint with a time-to-collision lower bound,* $\mathrm{TTC}_t \geq t_{\min}$*, whose gradient with respect to* a_t *is non-zero and admits a valid one-dimensional repair.*
(c) Predictive entry barrier*: add a one-step lookahead check that refuses to enter states from which* $\phi(x_{t+1}) =$ False *is unavoidable, reducing the observed TSVR to the T3 envelope.*

Without one of these routes, the AV residual is a structural *residual, not a conformal-coverage residual.*

a) Current closure route.: This dissertation adopts the TTC reformulation together with the predictive entry barrier as the canonical AV promotion path. The two-dimensional (a, v) repair remains a valid future route, but it is not the route claimed here. Under the bounded TTC contract, the closure matrix now records the AV row as proof-validated: locked replay reduces TSVR to zero on the canonical validation harness, the conservative entry-barrier soundness cases clear, and the CertOS lifecycle executes on the same adapter surface. This promotion is deliberately narrow. It does not promote every richer headway, lane-change, or multi-agent vehicle setting to equal proof depth.

G. Chapter Summary

This chapter closes the monograph at the correct control-theory level.

1) **T9** frames observation-aware safety logic as structurally necessary under persistent degradation.
2) **T10** turns mean reliability into a quantitative infrastructure-to-risk frontier.
3) **T11** packages universality as a typed kernel and adapter template, not as a claim that every domain already has battery-level proof closure.
4) **The unification theorem** explains how prior safe-control families sit inside the same degraded-observation framework.

The central theorem contribution is now precise: the safe-control object under degraded telemetry is not the nominal state alone but a reliability-indexed observation-consistent state set together with a repair map and a causal certificate surface. Battery supplies the full proof witness. T9 and T10 show that degraded observation creates a necessity and a

lower bound. T11 shows how the same kernel transfers when a new domain discharges the explicit structural obligations. This is a defensible control-theory claim; it is not an assertion that all evaluated domains already enjoy equal proof closure.

X. Integrated Theorem Surface Register

This appendix audits the full integrated theorem-like surface of the thesis rather than only the flagship battery ladder. The T1–T8 ladder in Chapters IV–VIII remains the canonical proof spine for the battery safety argument, and Chapter IX adds the later T9–T11 extension. Together they sit inside a broader integrated surface that also includes the precursor results in Part I, supporting lemmas, propositions, corollaries, definitions, and the standing assumption register.

The register below is generated directly from the LaTeX source files by `scripts/build_theorem_surface_register.py`. It is intended to answer a simple governance question without ambiguity: which theorem-like statements are present in the active thesis, where do they live, and which ones belong to the defended unique theorem program as opposed to the supporting integrated surface. The older hard-gated 18-row release status is reported in Appendix I; this appendix records the full inventory that the release gate audits.

The surface counts are summarised in Table VIII (Chapter III).

TABLE XXII: Integrated theorem-like surface register generated from the active thesis include tree.

ID	Group	Kind	Title	Context	Source
Defended theorem program (17; theorem:17)					
S1	Defended theorem program	theorem	Existence of the illusion under dropout	Formal Statement of the Illusion	`chapters_merged/ch03_background.tex`:495
S2	Defended theorem program	theorem	DC3S Feasibility Guarantee	Why the Illusion Demands a Runtime Shield	`chapters_merged/ch03_background.tex`:658
T1	Defended theorem program	theorem	OASG Existence — General CPS	Domain-Agnostic Statement	`chapters_merged/ch04_theoretical_foundations.tex`:681
T1	Defended theorem program	theorem	OASG Existence	Main Theorem: OASG Existence	`chapters_merged/ch04_theoretical_foundations.tex`:800
T2	Defended theorem program	theorem	Safety Preservation — General CPS	Domain-Agnostic Statement	`chapters_merged/ch04_theoretical_foundations.tex`:930
T2	Defended theorem program	theorem	Safety Preservation	Main Theorem: Safety Preservation	`chapters_merged/ch04_theoretical_foundations.tex`:1039
T3	Defended theorem program	theorem	ORIUS Core Bound — General CPS	Domain-Agnostic Statement	`chapters_merged/ch04_theoretical_foundations.tex`:1186
T3	Defended theorem program	theorem	ORIUS Core Bound	Main Theorem: ORIUS Core Bound	`chapters_merged/ch04_theoretical_foundations.tex`:1274

Continued on next page

TABLE XXII — continued

ID	Group	Kind	Title	Context	Source
T4	Defended theorem program	theorem	No Free Safety — General CPS	Domain-Agnostic Statement	chapters_merged/ch04_theoretical_foundations.tex:1431
T4	Defended theorem program	theorem	No Free Safety	Main Theorem: No Free Safety	chapters_merged/ch04_theoretical_foundations.tex:1537
T5	Defended theorem program	theorem	Certificate validity horizon	Main Temporal Theorems	chapters_merged/ch04_theoretical_foundations.tex:1739
T6	Defended theorem program	theorem	Certificate expiration bound	Main Temporal Theorems	chapters_merged/ch04_theoretical_foundations.tex:1769
T7	Defended theorem program	theorem	Feasible fallback existence	Fallback and Graceful Degradation	chapters_merged/ch04_theoretical_foundations.tex:1848
T8	Defended theorem program	theorem	Graceful degradation dominance	Fallback and Graceful Degradation	chapters_merged/ch04_theoretical_foundations.tex:1874
T9	Defended theorem program	theorem	Universal Impossibility, T9	Observation-Aware Control Is Structurally Necessary (T9)	chapters/ch37_universality_completeness.tex:109
T10	Defended theorem program	theorem	Boundary-indistinguishability lower bound, T10	Boundary-Indistinguishability Lower Bound (T10)	chapters/ch37_universality_completeness.tex:206
T11	Defended theorem program	theorem	Typed structural transfer theorem, T11	Typed Structural Transfer and Failure-Mode Converse (T11)	chapters/ch37_universality_completeness.tex:338
Supporting chapter result (43; theorem:1, lemma:5, proposition:8, corollary:11, definition:15, remark:3)					
—	Supporting chapter result	definition	Observation-Action Safety Gap (OASG)	The Observation-Action Safety Gap	chapters_merged/ch01_introduction.tex:79
—	Supporting chapter result	definition	ORIUS-MDP	ORIUS-MDP Formalisation	chapters_merged/ch01_introduction.tex:315
—	Supporting chapter result	remark	—	ORIUS-MDP Formalisation	chapters_merged/ch01_introduction.tex:340
—	Supporting chapter result	definition	Battery-domain OASG	The Hidden Safety Gap	chapters_merged/ch01_introduction.tex:353

Continued on next page

TABLE XXII — continued

ID	Group	Kind	Title	Context	Source
—	Supporting chapter result	definition	Observed-State Safety Illusion	Formal Statement of the Illusion	chapters_merged/ch03_background.tex:472
—	Supporting chapter result	proposition	Insufficiency of observed-state evaluation	Why Standard Evaluation Misses the Problem	chapters_merged/ch03_background.tex:568
—	Supporting chapter result	definition	True-State Violation Rate (TSVR)	CPSBench and Dual-Trajectory Evaluation	chapters_merged/ch03_background.tex:606
—	Supporting chapter result	definition	Intervention Rate (IR)	CPSBench and Dual-Trajectory Evaluation	chapters_merged/ch03_background.tex:618
—	Supporting chapter result	definition	Abstract CPS	Domain-Agnostic Statement	chapters_merged/ch04_theoretical_foundations.tex:668
—	Supporting chapter result	definition	Observed safety	Observed and True Safety Predicates	chapters_merged/ch04_theoretical_foundations.tex:724
—	Supporting chapter result	definition	True safety	Observed and True Safety Predicates	chapters_merged/ch04_theoretical_foundations.tex:737
—	Supporting chapter result	lemma	Observation gap under dropout	Auxiliary Lemmas	chapters_merged/ch04_theoretical_foundations.tex:766
—	Supporting chapter result	lemma	Boundary proximity under arbitrage	Auxiliary Lemmas	chapters_merged/ch04_theoretical_foundations.tex:784
—	Supporting chapter result	corollary	OASG rate lower bound	Corollaries	chapters_merged/ch04_theoretical_foundations.tex:875
—	Supporting chapter result	corollary	OASG severity	Corollaries	chapters_merged/ch04_theoretical_foundations.tex:883
—	Supporting chapter result	definition	Tightened safe-action set	Battery-Domain Instantiation: Tightened Safe Action Set	chapters_merged/ch04_theoretical_foundations.tex:966
—	Supporting chapter result	proposition	Inflated set contains the current state	Auxiliary Propositions	chapters_merged/ch04_theoretical_foundations.tex:1000
—	Supporting chapter result	proposition	Tightened feasibility implies true feasibility	Auxiliary Propositions	chapters_merged/ch04_theoretical_foundations.tex:1017

Continued on next page

TABLE XXII — continued

ID	Group	Kind	Title	Context	Source
—	Supporting chapter result	corollary	Zero-violation regime	Corollaries	chapters_merged/ch04_theoretical_foundations.tex:1121
—	Supporting chapter result	corollary	Intervention-safety tradeoff	Corollaries	chapters_merged/ch04_theoretical_foundations.tex:1128
—	Supporting chapter result	lemma	Aggregation under a predictable risk budget	Theorem-Local Risk-Envelope Contract	chapters_merged/ch04_theoretical_foundations.tex:1246
—	Supporting chapter result	corollary	Perfect-telemetry collapse	Corollaries	chapters_merged/ch04_theoretical_foundations.tex:1358
—	Supporting chapter result	corollary	Reliability-proportional safety	Corollaries	chapters_merged/ch04_theoretical_foundations.tex:1366
—	Supporting chapter result	corollary	Intervention sufficiency	Corollaries	chapters_merged/ch04_theoretical_foundations.tex:1374
—	Supporting chapter result	definition	Quality-ignorant controller	Battery-Domain Instantiation: Quality-Ignorant Controller Class	chapters_merged/ch04_theoretical_foundations.tex:1457
—	Supporting chapter result	lemma	Admissible fault sequence existence	Auxiliary Lemmas	chapters_merged/ch04_theoretical_foundations.tex:1476
—	Supporting chapter result	lemma	No margin compensation for quality-ignorant controllers	Auxiliary Lemmas	chapters_merged/ch04_theoretical_foundations.tex:1498
—	Supporting chapter result	corollary	Reliability awareness is necessary	Corollary	chapters_merged/ch04_theoretical_foundations.tex:1608
—	Supporting chapter result	definition	Safety certificate	Certificate Object and Validity Horizon	chapters_merged/ch04_theoretical_foundations.tex:1691
—	Supporting chapter result	definition	Forward tube	Forward-Tube Construction	chapters_merged/ch04_theoretical_foundations.tex:1718
—	Supporting chapter result	corollary	Certificate half-life	Main Temporal Theorems	chapters_merged/ch04_theoretical_foundations.tex:1791

Continued on next page

TABLE XXII — continued

ID	Group	Kind	Title	Context	Source
—	Supporting chapter result	proposition	Safe-budget monotonicity	Fallback and Graceful Degradation	chapters_merged/ch04_theoretical_foundations.tex:1905
—	Supporting chapter result	proposition	Conditional conservatism	Intervention Rate and Conditional Conservatism	chapters_merged/ch06_main_results.tex:82
—	Supporting chapter result	proposition	Intervention lead time	Intervention Timing	chapters_merged/ch06_main_results.tex:591
—	Supporting chapter result	definition	Degraded-observation safe-control problem	Universal Problem Statement and Information Pattern	chapters/ch37_universality_completeness.tex:58
—	Supporting chapter result	remark	—	Observation-Aware Control Is Structurally Necessary (T9)	chapters/ch37_universality_completeness.tex:175
—	Supporting chapter result	remark	—	Boundary-Indistinguishability Lower Bound (T10)	chapters/ch37_universality_completeness.tex:266
—	Supporting chapter result	definition	Universal adapter contract	Typed Structural Transfer and Failure-Mode Converse (T11)	chapters/ch37_universality_completeness.tex:300
—	Supporting chapter result	corollary	Episode aggregation under explicit per-step budgets	Typed Structural Transfer and Failure-Mode Converse (T11)	chapters/ch37_universality_completeness.tex:385
—	Supporting chapter result	proposition	Failure of any transfer obligation breaks the reference proof pattern	Typed Structural Transfer and Failure-Mode Converse (T11)	chapters/ch37_universality_completeness.tex:417
—	Supporting chapter result	theorem	Unification	Unification: ORIUS Subsumes Prior Frameworks	chapters/ch37_universality_completeness.tex:486
—	Supporting chapter result	proposition	Constraint-class mismatch barrier	Constraint-Class Mismatch: When 1-D Action Repair Is Insufficient	chapters/ch37_universality_completeness.tex:548
—	Supporting chapter result	corollary	AV promotion routes from the original headway mismatch	Constraint-Class Mismatch: When 1-D Action Repair Is Insufficient	chapters/ch37_universality_completeness.tex:598
Appendix proof restatement (10; theorem:8, proposition:1, corollary:1)					
—	Appendix proof restatement	theorem	C.1 Battery-Domain OASG Existence	C.1 Battery-Domain OASG Existence	appendices/app_c_full_proofs.tex:32
—	Appendix proof restatement	theorem	C.2 One-Step Safety Preservation	C.2 One-Step Safety Preservation	appendices/app_c_full_proofs.tex:45

Continued on next page

TABLE XXII — continued

ID	Group	Kind	Title	Context	Source
—	Appendix proof restatement	theorem	C.3 Battery ORIUS Core Bound	C.3 Battery ORIUS Core Bound	appendices/app_c_full_proofs.tex:99
—	Appendix proof restatement	theorem	C.4 No Free Safety	C.4 No Free Safety	appendices/app_c_full_proofs.tex:156
—	Appendix proof restatement	theorem	C.5 Certificate Validity Horizon	C.5 Certificate Validity Horizon	appendices/app_c_full_proofs.tex:185
—	Appendix proof restatement	theorem	C.6 Certificate Expiration Bound	C.6 Certificate Expiration Bound	appendices/app_c_full_proofs.tex:200
—	Appendix proof restatement	corollary	C.7 Certificate Half-Life Corollary	C.7 Certificate Half-Life Corollary	appendices/app_c_full_proofs.tex:216
—	Appendix proof restatement	theorem	C.8 Feasible Fallback Existence	C.8 Feasible Fallback Existence	appendices/app_c_full_proofs.tex:228
—	Appendix proof restatement	theorem	C.9 Graceful Degradation Dominance	C.9 Graceful Degradation Dominance	appendices/app_c_full_proofs.tex:241
—	Appendix proof restatement	proposition	C.10 Safe-Budget Monotonicity Proposition	C.10 Safe-Budget Monotonicity Proposition	appendices/app_c_full_proofs.tex:257
Assumption register (16; assumption:16)					
A1	Assumption register	assumption	A1: Bounded model error	Assumption IDs (A1–A8)	chapters_merged/ch04_theoretical_foundations.tex:415
A2	Assumption register	assumption	A2: Bounded telemetry error	Assumption IDs (A1–A8)	chapters_merged/ch04_theoretical_foundations.tex:422
A3	Assumption register	assumption	A3: Feasible safe repair	Assumption IDs (A1–A8)	chapters_merged/ch04_theoretical_foundations.tex:429
A4	Assumption register	assumption	A4: Known dynamics	Assumption IDs (A1–A8)	chapters_merged/ch04_theoretical_foundations.tex:435
A5	Assumption register	assumption	A5: Monotone bounded uncertainty inflation	Assumption IDs (A1–A8)	chapters_merged/ch04_theoretical_foundations.tex:443
A6	Assumption register	assumption	A6: Bounded detector lag	Assumption IDs (A1–A8)	chapters_merged/ch04_theoretical_foundations.tex:450

Continued on next page

TABLE XXII — continued

ID	Group	Kind	Title	Context	Source
A7	Assumption register	assumption	A7: Causal certificate update	Assumption IDs (A1–A8)	chapters_merged/ch04_theoretical_foundations.tex:469
A8	Assumption register	assumption	A8: Admissible fallback policy	Assumption IDs (A1–A8)	chapters_merged/ch04_theoretical_foundations.tex:475
A1	Assumption register	assumption	A1 — Bounded model error	Detailed Statements	appendices/app_b_assumptions.tex:47
A2	Assumption register	assumption	A2 — Bounded telemetry error	Detailed Statements	appendices/app_b_assumptions.tex:52
A3	Assumption register	assumption	A3 — Feasible safe repair	Detailed Statements	appendices/app_b_assumptions.tex:57
A4	Assumption register	assumption	A4 — Known dynamics	Detailed Statements	appendices/app_b_assumptions.tex:62
A5	Assumption register	assumption	A5 — Monotone bounded inflation	Detailed Statements	appendices/app_b_assumptions.tex:67
A6	Assumption register	assumption	A6 — Bounded detector lag	Detailed Statements	appendices/app_b_assumptions.tex:72
A7	Assumption register	assumption	A7 — Causal certificate	Detailed Statements	appendices/app_b_assumptions.tex:77
A8	Assumption register	assumption	A8 — Admissible fallback	Detailed Statements	appendices/app_b_assumptions.tex:82

XI. Formula, Symbol, and Term Register

This appendix collects the canonical vocabulary of the monograph. It is intentionally redundant: repetition here prevents terminology drift in the main text.

TABLE XXIII: Formula, symbol, and term register for the ORIUS monograph.

Symbol / Term	Meaning	Role in the monograph
OASG	Observation-action safety gap	The event where an action is admissible on observed state but unsafe on true state.
True-state violation	Physical safety failure on latent plant state	Primary universal benchmark outcome.
Observed-state satisfaction	Legality under the sensed or estimated state	Lets ORIUS separate nominal legality from physical legality.
Constraint margin	Distance to the unsafe set	Shared quantity used to compare action safety across domains.
Repair	Projection of the candidate action into the tightened admissible set	Core shield operation in ORIUS.

Symbol / Term	Meaning	Role in the monograph
Fallback	Replacement action emitted when no admissible repaired action exists	Safety-preserving terminal policy in degraded conditions.
Certificate	Structured runtime evidence object attached to a released action	Connects technical safety to governance.
Reliability weight	Scalar trust measure over the observation channel	Controls uncertainty widening and intervention severity.
Inflation	Widening of the observation-consistent state set	Makes degraded telemetry visible to the action layer.
CertOS	Runtime governance layer	Tracks validity, lifecycle, fallback, and audit continuity.
Empirical witness	Deepest evidence tier	Current battery status in the monograph.
Bounded proof surface	Operational domain with verified improvement but bounded scope	Current industrial and healthcare status.
Promotion candidate	Working domain with unresolved closure gap	Reserved for future rows that approach defended promotion without yet clearing the parity gate.
Shadow-synthetic	Portability row with structural compatibility but weaker empirical closure	Current navigation status.
Experimental	Early or placeholder domain surface	Current aerospace status.
Term 16	Extended monograph glossary entry	Used to keep universal chapters precise and to avoid silent drift back into battery-specific language.
Term 17	Extended monograph glossary entry	Used to keep universal chapters precise and to avoid silent drift back into battery-specific language.
Term 18	Extended monograph glossary entry	Used to keep universal chapters precise and to avoid silent drift back into battery-specific language.
Term 19	Extended monograph glossary entry	Used to keep universal chapters precise and to avoid silent drift back into battery-specific language.
Term 20	Extended monograph glossary entry	Used to keep universal chapters precise and to avoid silent drift back into battery-specific language.
Term 21	Extended monograph glossary entry	Used to keep universal chapters precise and to avoid silent drift back into battery-specific language.
Term 22	Extended monograph glossary entry	Used to keep universal chapters precise and to avoid silent drift back into battery-specific language.
Term 23	Extended monograph glossary entry	Used to keep universal chapters precise and to avoid silent drift back into battery-specific language.
Term 24	Extended monograph glossary entry	Used to keep universal chapters precise and to avoid silent drift back into battery-specific language.
Term 25	Extended monograph glossary entry	Used to keep universal chapters precise and to avoid silent drift back into battery-specific language.
Term 26	Extended monograph glossary entry	Used to keep universal chapters precise and to avoid silent drift back into battery-specific language.
Term 27	Extended monograph glossary entry	Used to keep universal chapters precise and to avoid silent drift back into battery-specific language.
Term 28	Extended monograph glossary entry	Used to keep universal chapters precise and to avoid silent drift back into battery-specific language.
Term 29	Extended monograph glossary entry	Used to keep universal chapters precise and to avoid silent drift back into battery-specific language.
Term 30	Extended monograph glossary entry	Used to keep universal chapters precise and to avoid silent drift back into battery-specific language.
Term 31	Extended monograph glossary entry	Used to keep universal chapters precise and to avoid silent drift back into battery-specific language.

Symbol / Term	Meaning	Role in the monograph
Term 32	Extended monograph glossary entry	Used to keep universal chapters precise and to avoid silent drift back into battery-specific language.
Term 33	Extended monograph glossary entry	Used to keep universal chapters precise and to avoid silent drift back into battery-specific language.
Term 34	Extended monograph glossary entry	Used to keep universal chapters precise and to avoid silent drift back into battery-specific language.
Term 35	Extended monograph glossary entry	Used to keep universal chapters precise and to avoid silent drift back into battery-specific language.
Term 36	Extended monograph glossary entry	Used to keep universal chapters precise and to avoid silent drift back into battery-specific language.
Term 37	Extended monograph glossary entry	Used to keep universal chapters precise and to avoid silent drift back into battery-specific language.
Term 38	Extended monograph glossary entry	Used to keep universal chapters precise and to avoid silent drift back into battery-specific language.
Term 39	Extended monograph glossary entry	Used to keep universal chapters precise and to avoid silent drift back into battery-specific language.
Term 40	Extended monograph glossary entry	Used to keep universal chapters precise and to avoid silent drift back into battery-specific language.
Term 41	Extended monograph glossary entry	Used to keep universal chapters precise and to avoid silent drift back into battery-specific language.
Term 42	Extended monograph glossary entry	Used to keep universal chapters precise and to avoid silent drift back into battery-specific language.
Term 43	Extended monograph glossary entry	Used to keep universal chapters precise and to avoid silent drift back into battery-specific language.
Term 44	Extended monograph glossary entry	Used to keep universal chapters precise and to avoid silent drift back into battery-specific language.
Term 45	Extended monograph glossary entry	Used to keep universal chapters precise and to avoid silent drift back into battery-specific language.
Term 46	Extended monograph glossary entry	Used to keep universal chapters precise and to avoid silent drift back into battery-specific language.
Term 47	Extended monograph glossary entry	Used to keep universal chapters precise and to avoid silent drift back into battery-specific language.
Term 48	Extended monograph glossary entry	Used to keep universal chapters precise and to avoid silent drift back into battery-specific language.
Term 49	Extended monograph glossary entry	Used to keep universal chapters precise and to avoid silent drift back into battery-specific language.
Term 50	Extended monograph glossary entry	Used to keep universal chapters precise and to avoid silent drift back into battery-specific language.
Term 51	Extended monograph glossary entry	Used to keep universal chapters precise and to avoid silent drift back into battery-specific language.
Term 52	Extended monograph glossary entry	Used to keep universal chapters precise and to avoid silent drift back into battery-specific language.
Term 53	Extended monograph glossary entry	Used to keep universal chapters precise and to avoid silent drift back into battery-specific language.
Term 54	Extended monograph glossary entry	Used to keep universal chapters precise and to avoid silent drift back into battery-specific language.
Term 55	Extended monograph glossary entry	Used to keep universal chapters precise and to avoid silent drift back into battery-specific language.
Term 56	Extended monograph glossary entry	Used to keep universal chapters precise and to avoid silent drift back into battery-specific language.
Term 57	Extended monograph glossary entry	Used to keep universal chapters precise and to avoid silent drift back into battery-specific language.
Term 58	Extended monograph glossary entry	Used to keep universal chapters precise and to avoid silent drift back into battery-specific language.
Term 59	Extended monograph glossary entry	Used to keep universal chapters precise and to avoid silent drift back into battery-specific language.

Symbol / Term	Meaning	Role in the monograph
Term 60	Extended monograph glossary entry	Used to keep universal chapters precise and to avoid silent drift back into battery-specific language.

References